



US012397944B2

(12) **United States Patent**
Itagaki et al.

(10) **Patent No.:** **US 12,397,944 B2**
(45) **Date of Patent:** **Aug. 26, 2025**

(54) **BINDING MACHINE**

(56) **References Cited**

(71) Applicant: **MAX CO., LTD.**, Tokyo (JP)

U.S. PATENT DOCUMENTS

(72) Inventors: **Osamu Itagaki**, Tokyo (JP); **Nobutaka Tashima**, Tokyo (JP)

5,462,098 A 10/1995 Murakami
8,197,379 B1 6/2012 Yin
9,593,496 B2 * 3/2017 Coles B65B 13/285
2010/0147411 A1 * 6/2010 Kusakari B65B 13/025
72/324
2011/0056389 A1 * 3/2011 Neeser B65B 13/322
100/29

(73) Assignee: **Max Co., Ltd.**, Tokyo (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 178 days.

2013/0284475 A1 10/2013 Hirabayashi et al.
(Continued)

FOREIGN PATENT DOCUMENTS

(21) Appl. No.: **17/964,346**

JP S63-191719 A 8/1988
JP 2011-183506 A 9/2011
(Continued)

(22) Filed: **Oct. 12, 2022**

(65) **Prior Publication Data**

US 2023/0123279 A1 Apr. 20, 2023

OTHER PUBLICATIONS

WO 2014/194434A1, Amacker et al. Dec. 2014.*
(Continued)

(30) **Foreign Application Priority Data**

Oct. 20, 2021 (JP) 2021-171966

Primary Examiner — Edward T Tolan

(74) *Attorney, Agent, or Firm* — Banner & Witcoff, Ltd.

(51) **Int. Cl.**

B65B 13/18 (2006.01)

B65B 27/10 (2006.01)

E04G 21/12 (2006.01)

(52) **U.S. Cl.**

CPC **B65B 13/18** (2013.01); **B65B 13/187**
(2013.01); **B65B 27/10** (2013.01); **E04G 21/12**
(2013.01)

(58) **Field of Classification Search**

CPC E04G 21/12; E04G 21/123; B65B 13/185;
B65B 13/187; B65B 13/28; B65B 13/285;
B65B 13/18; B65B 27/10; B21F 7/00;
B21F 15/04

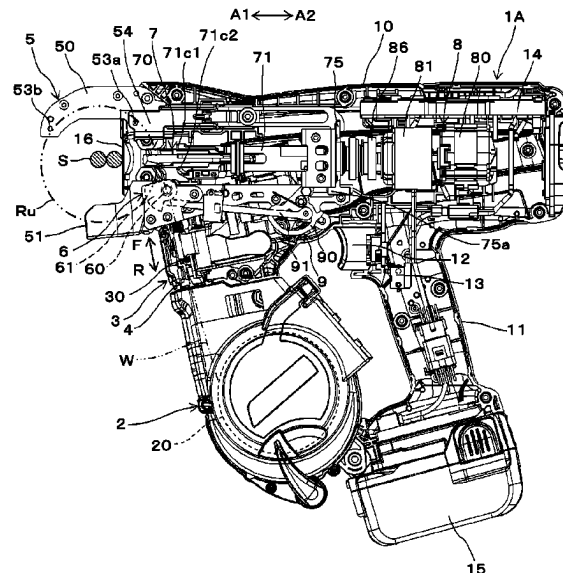
See application file for complete search history.

(57)

ABSTRACT

A binding machine includes a wire feeding unit that feeds a wire, a curl forming unit that forms a path along which the wire fed by the wire feeding unit is to be wound around an object, a cutting unit that cuts the wire wound on the object, a binding unit that twists the wire wound on the object and cut by the cutting unit, and a drive unit that drives the binding unit. The drive unit includes a motor and a decelerator that perform deceleration and amplification of torque. The decelerator includes a sun gear, a planetary gear in mesh with the sun gear, a planet cage that supports the planetary gear, and an internal gear in mesh with the planetary gear. The planet cage is rotatably supported on one side and the other side along an axis direction.

5 Claims, 19 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2015/0210411 A1* 7/2015 Finzo B65B 13/322
100/32
2023/0294859 A1* 9/2023 He B65B 13/187
140/93.6

FOREIGN PATENT DOCUMENTS

JP 2013-039628 A 2/2013
JP 2017-089270 A 5/2017
JP 2018-108850 A 7/2018
WO 2008/062213 A2 5/2008
WO 2017/082002 A1 5/2017
WO 2020-170297 A1 8/2020

OTHER PUBLICATIONS

Extended European Search Report dated Mar. 22, 2023, issued by the European Patent Office in the corresponding European Patent Application No. 22201911.9. (7 pages).

* cited by examiner

FIG. 1

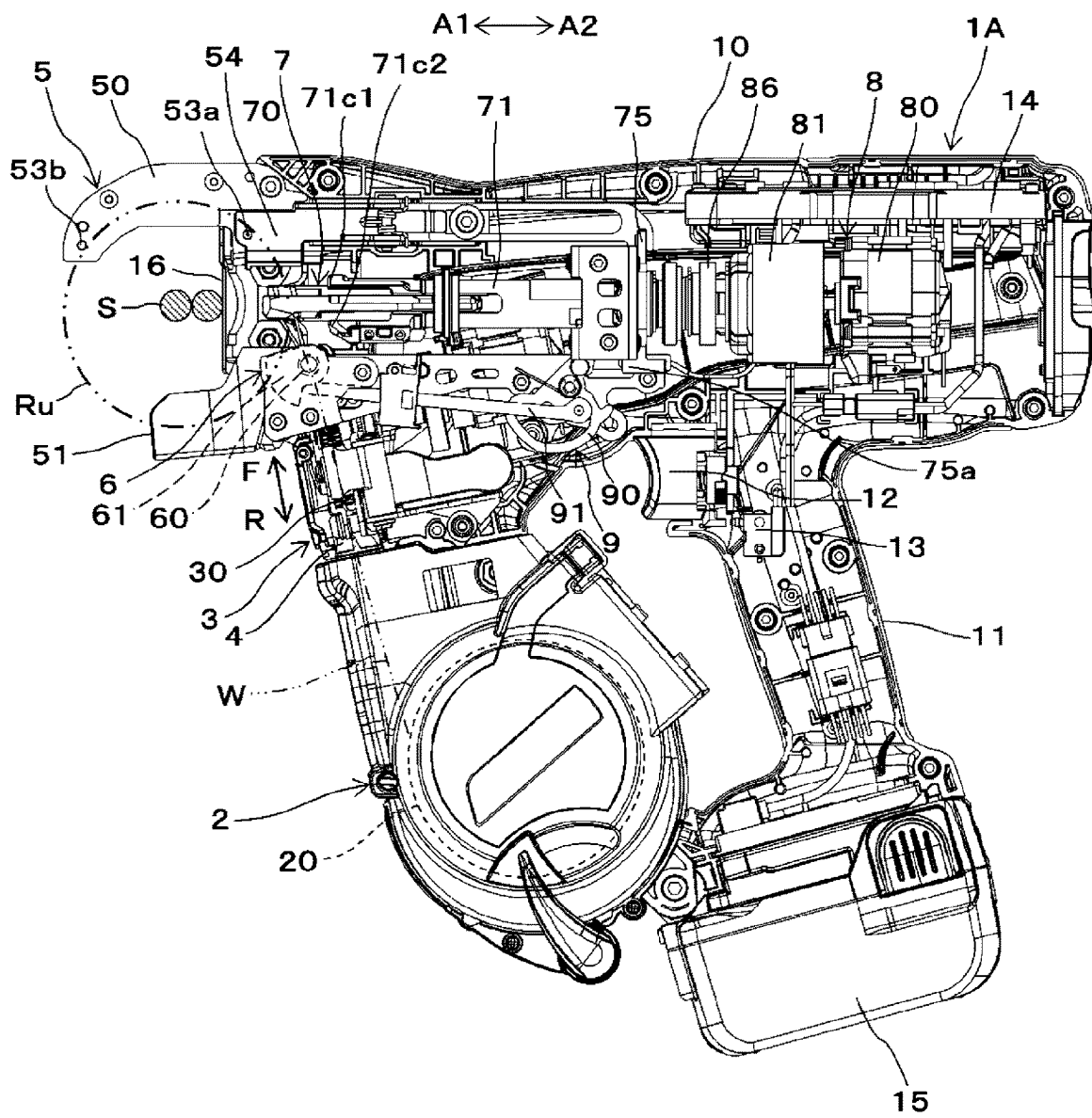


FIG. 2A

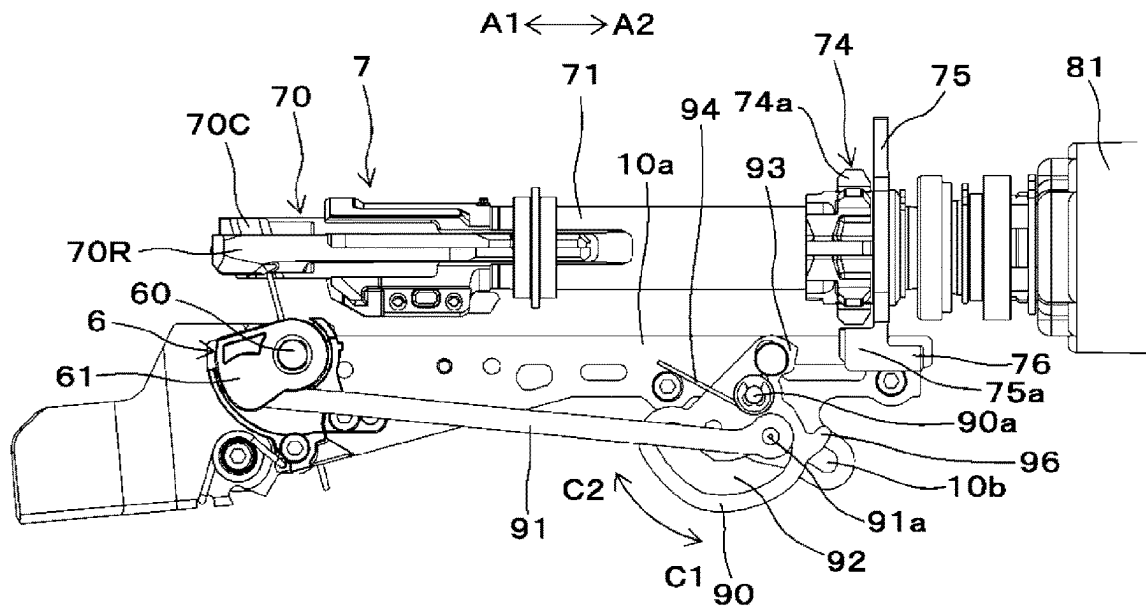


FIG. 2B

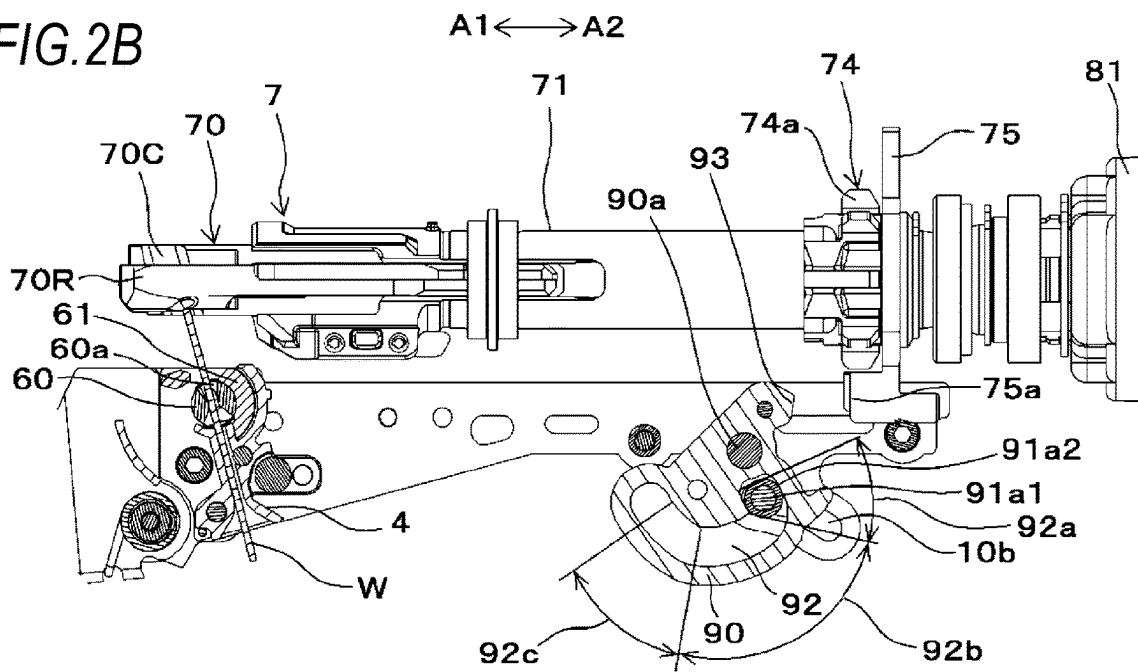


FIG.2C

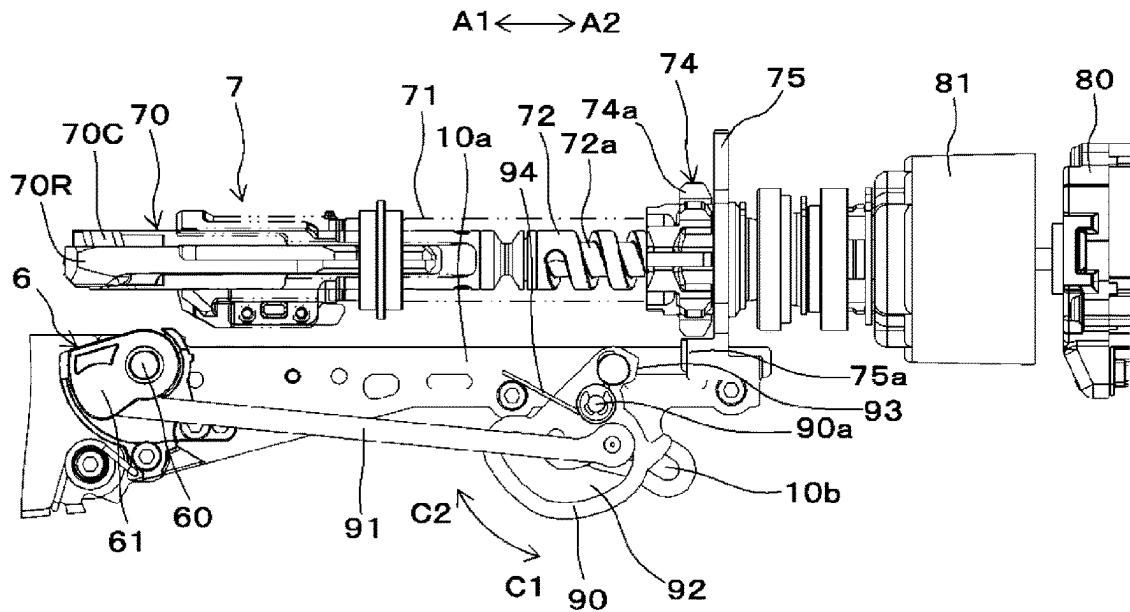


FIG.3A

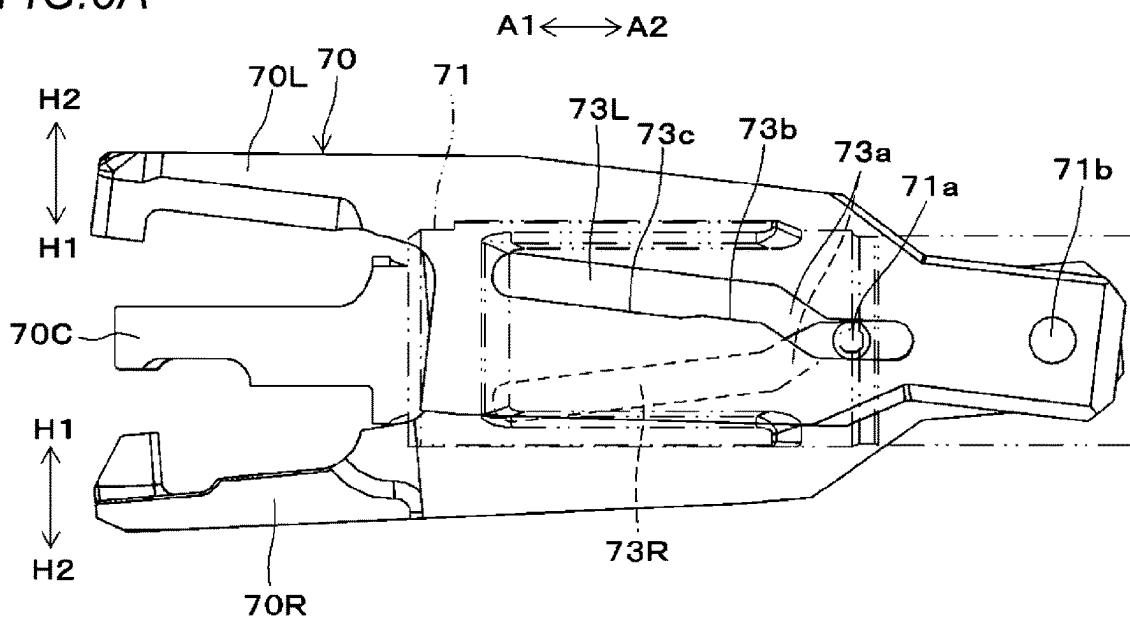


FIG.3B

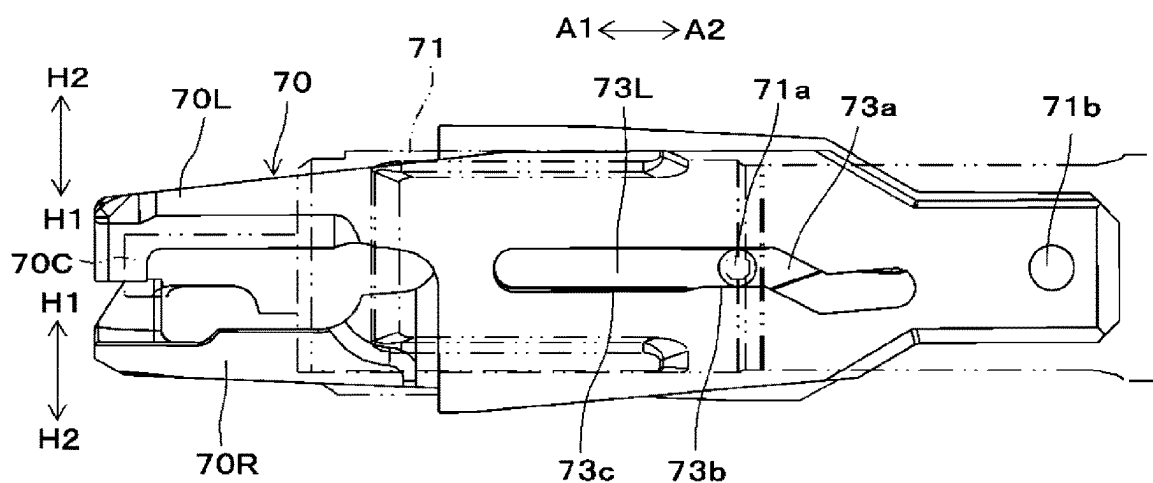


FIG.3C

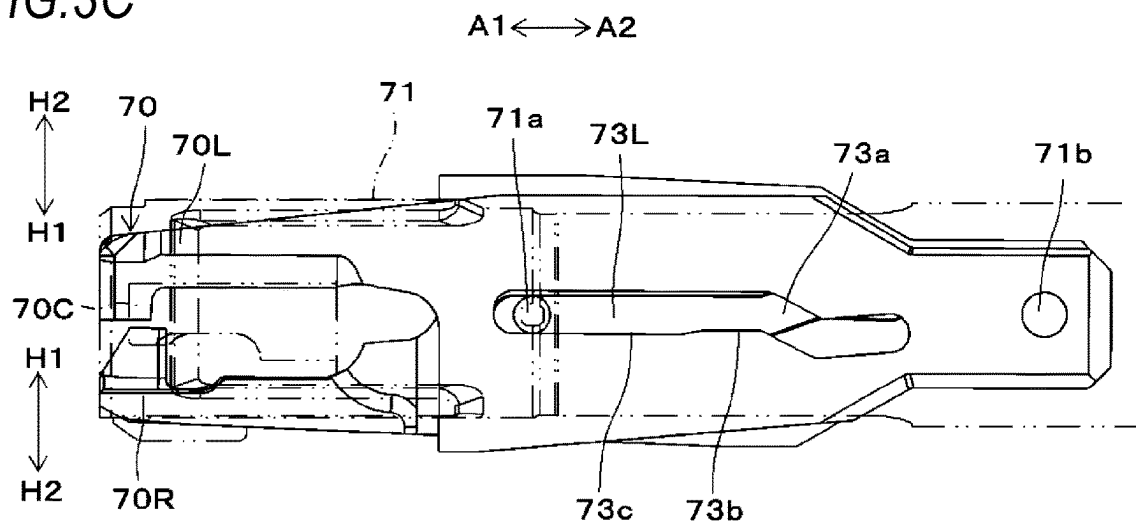


FIG.3D

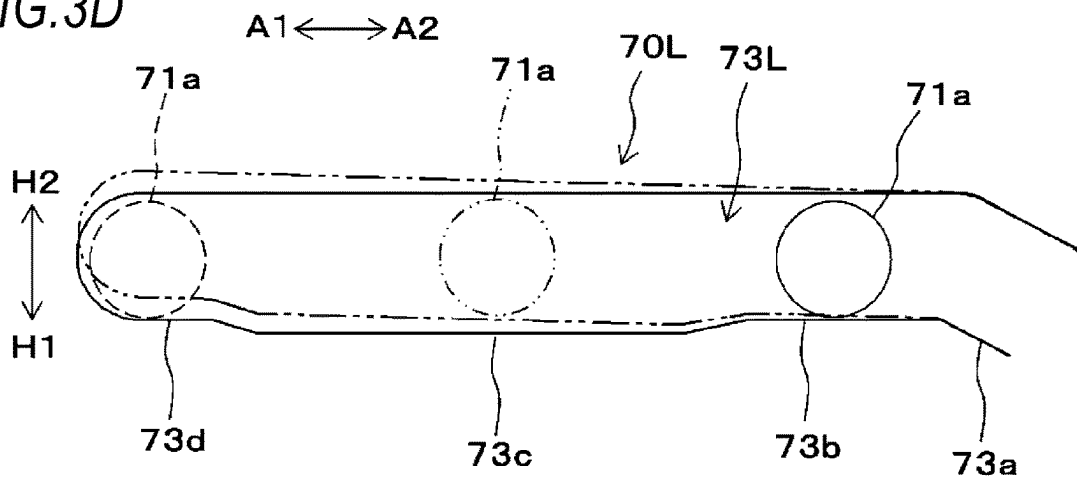


FIG. 3E

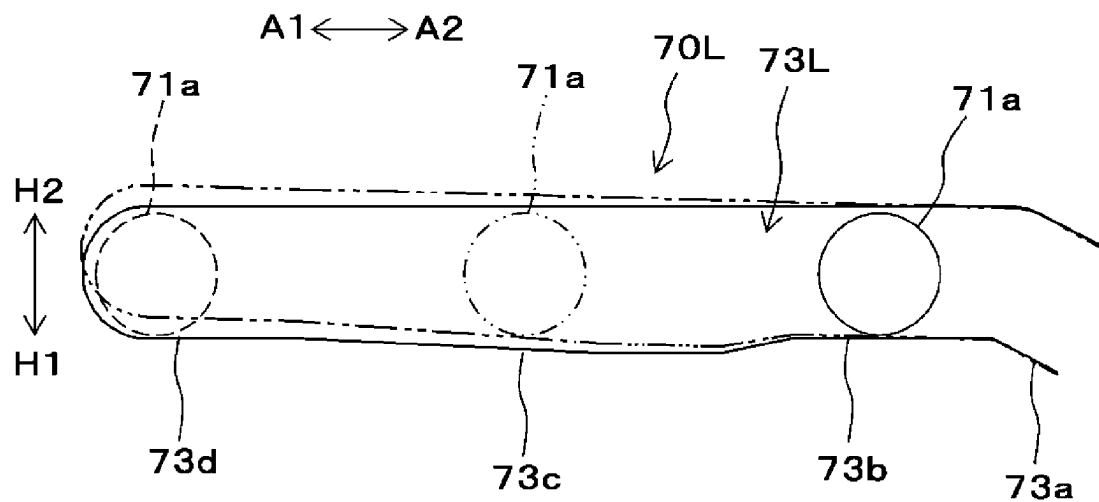


FIG. 3F

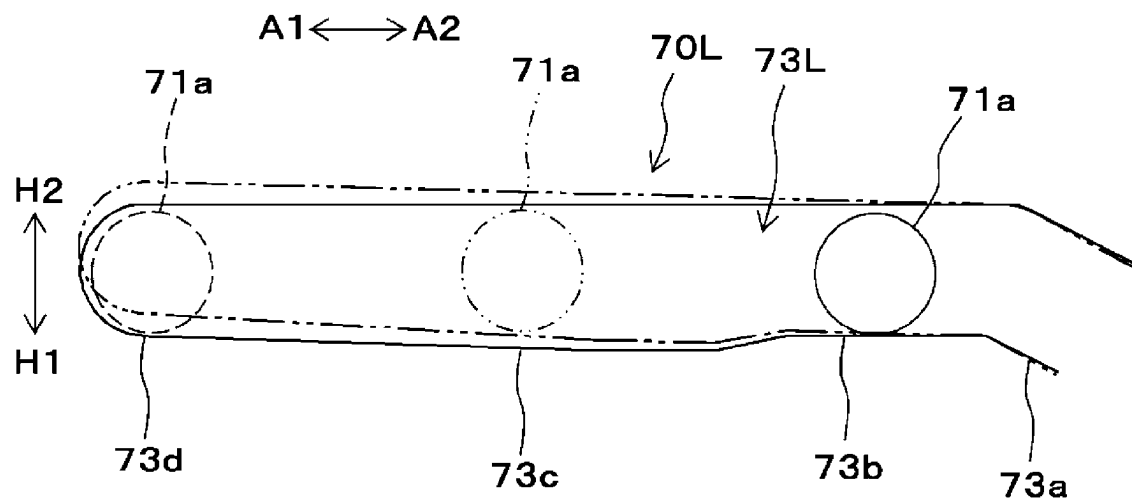


FIG. 4A

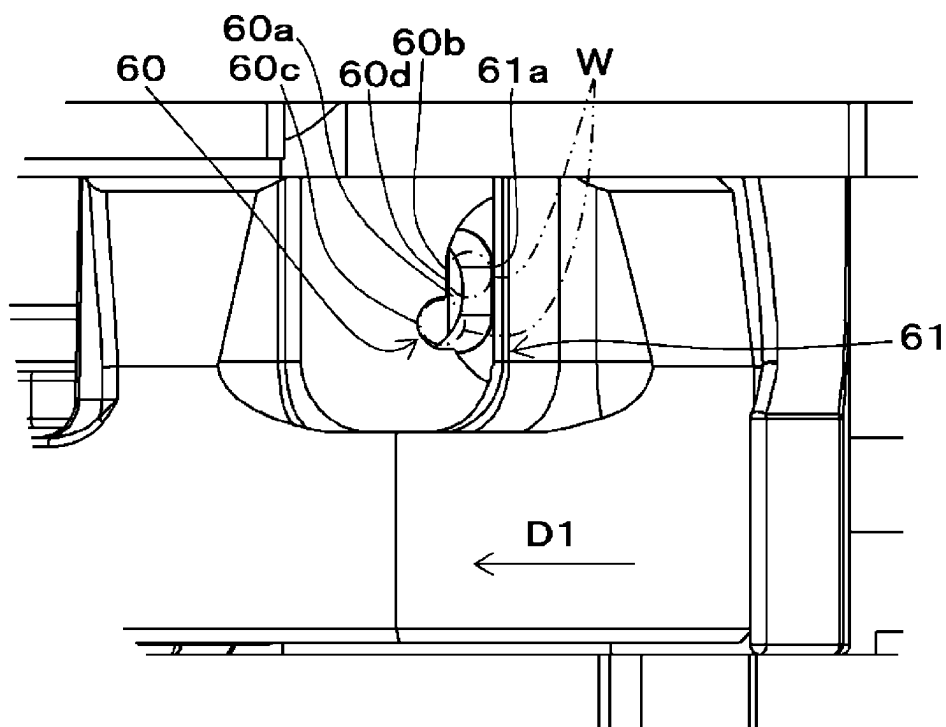


FIG. 4B

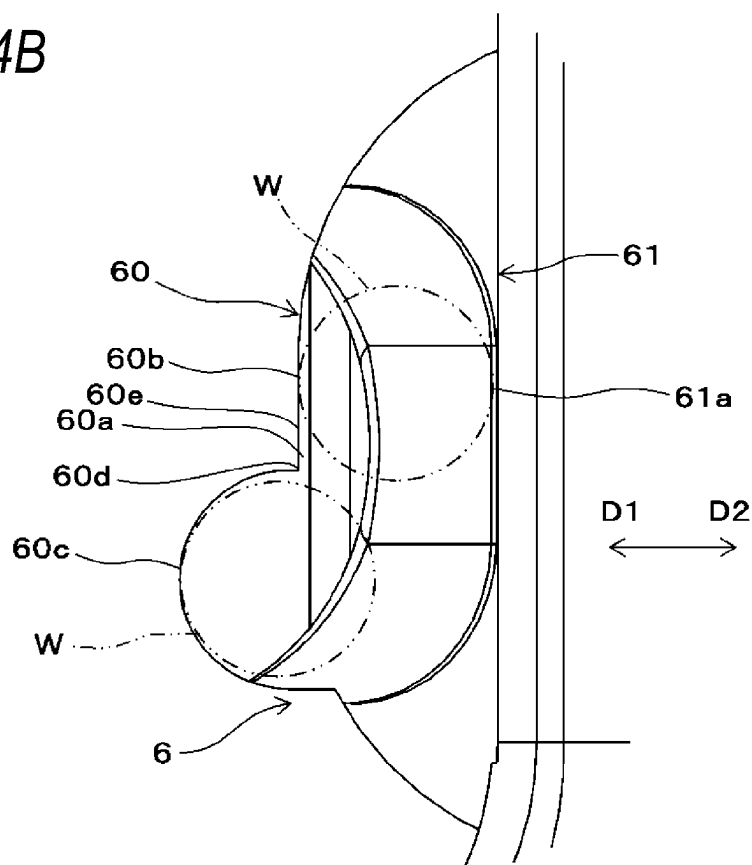


FIG.4C

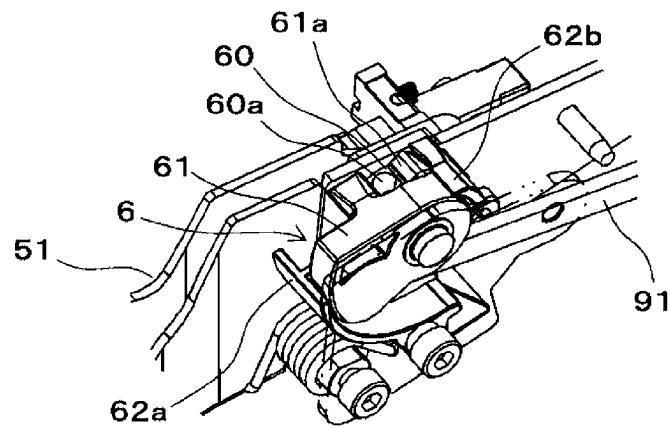


FIG.4D

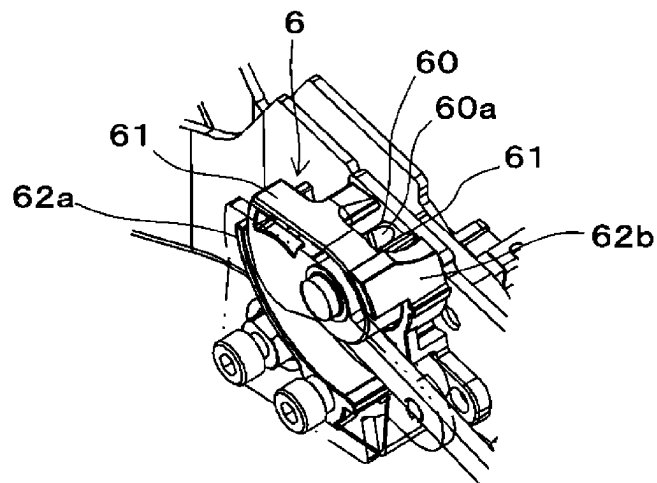


FIG.4E

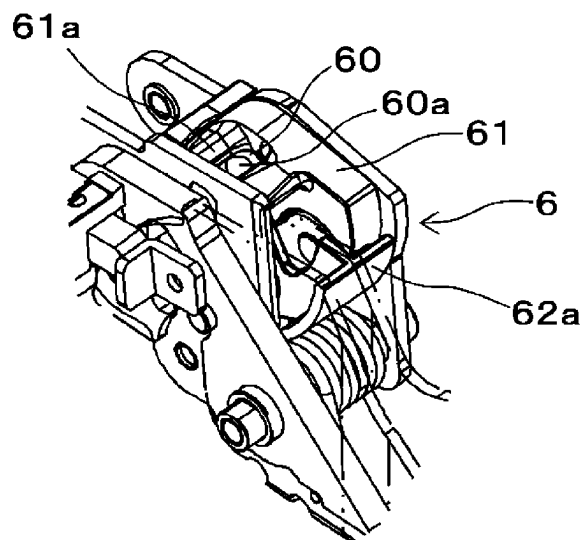


FIG. 4F

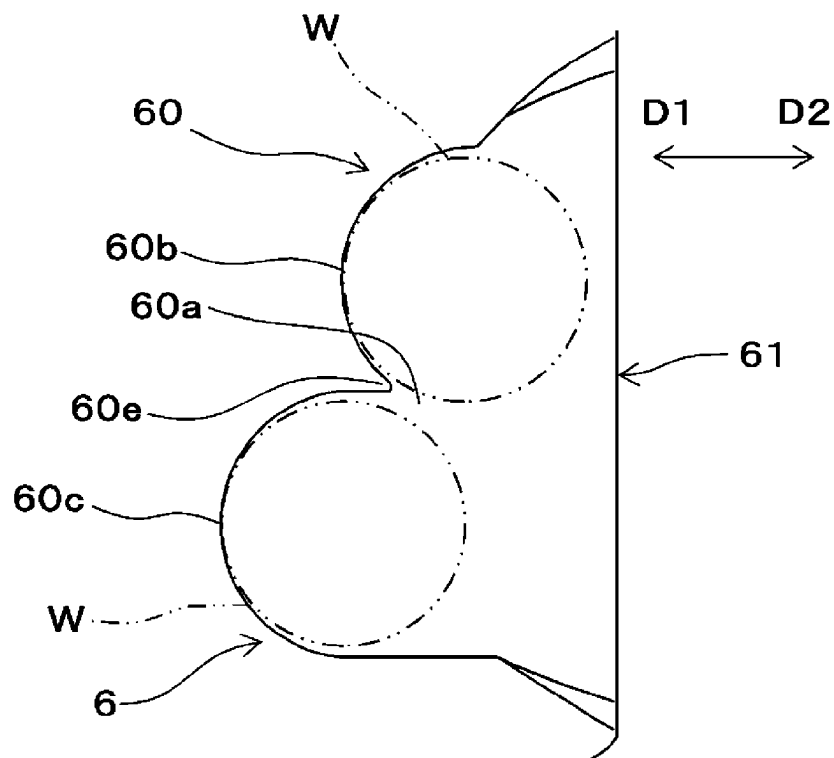


FIG. 4G

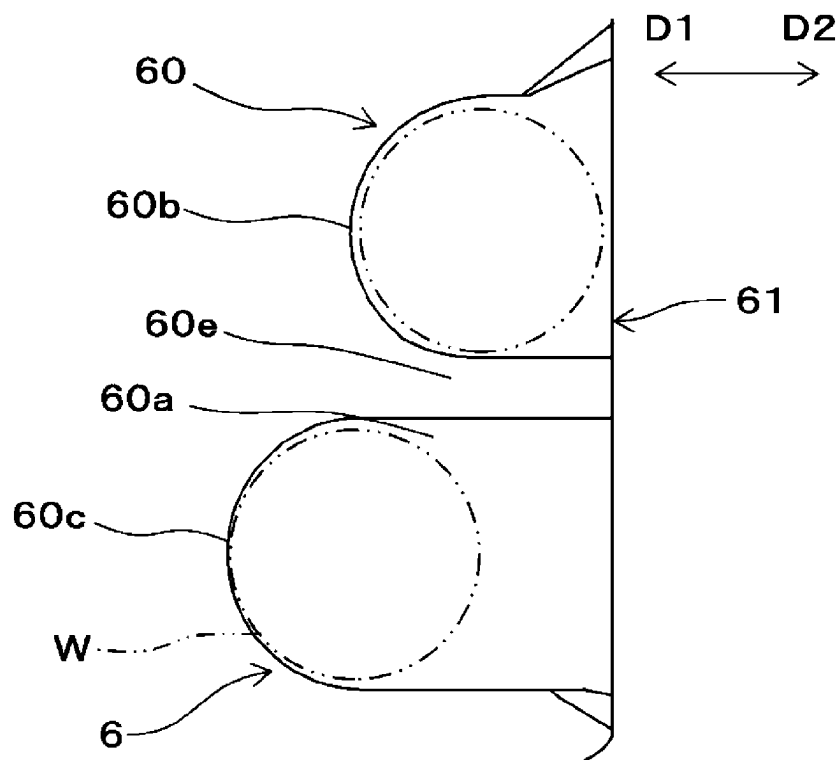


FIG. 5A

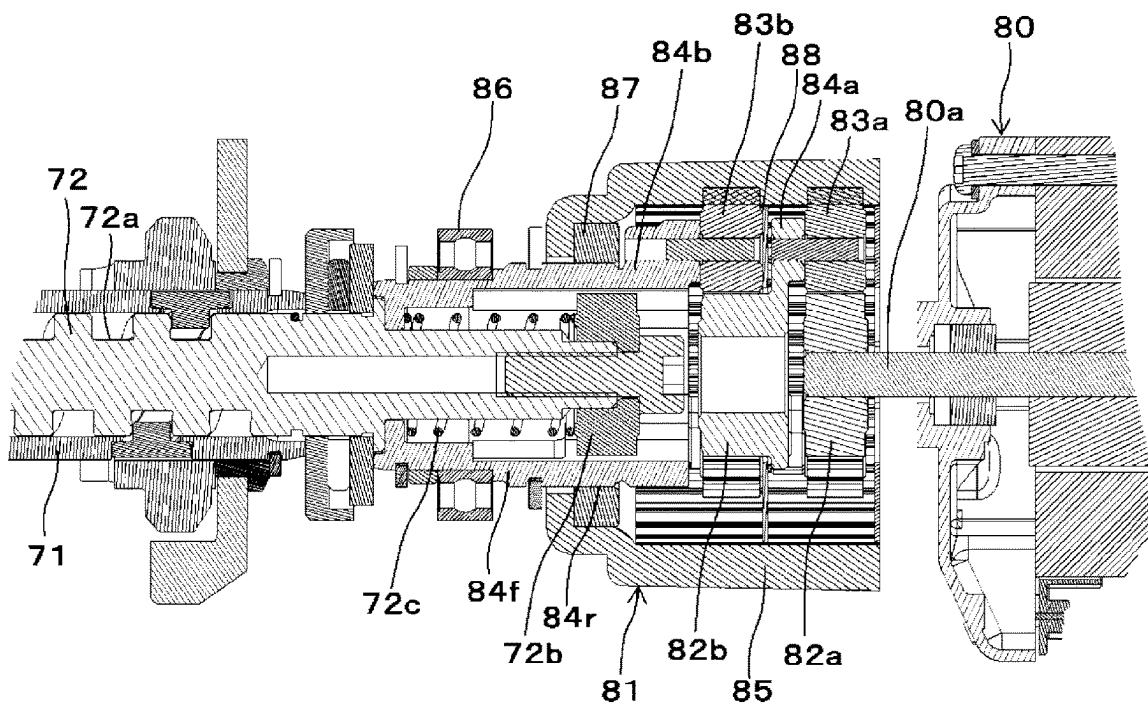


FIG. 5B

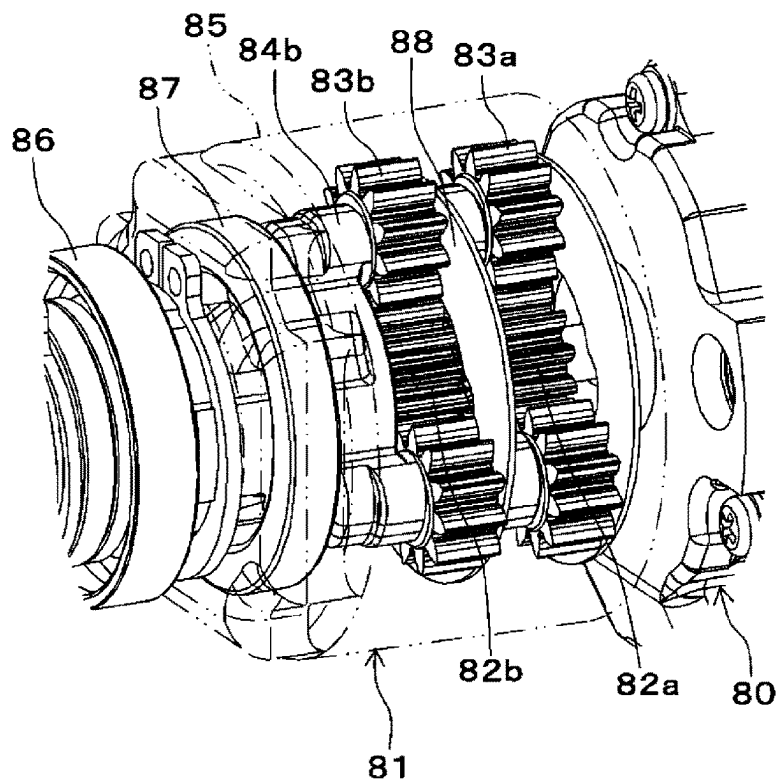


FIG. 5C

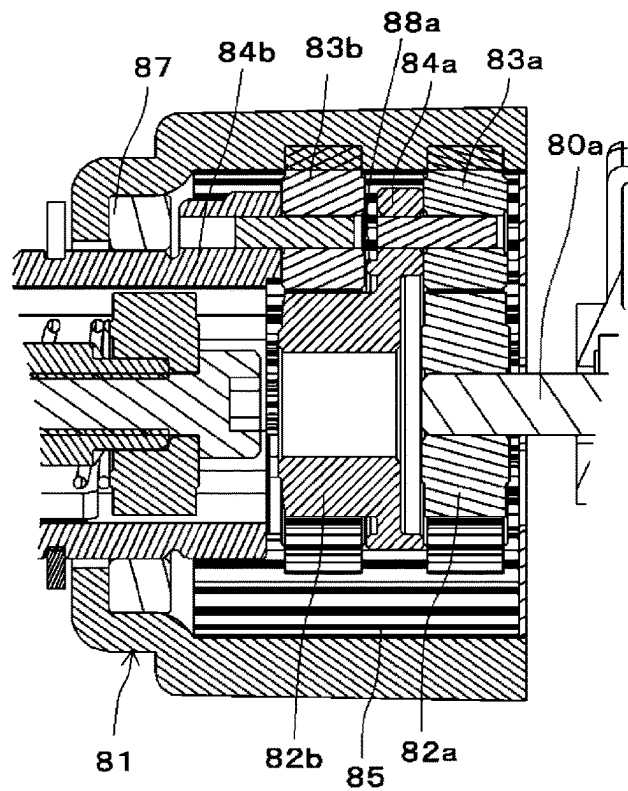


FIG. 5D

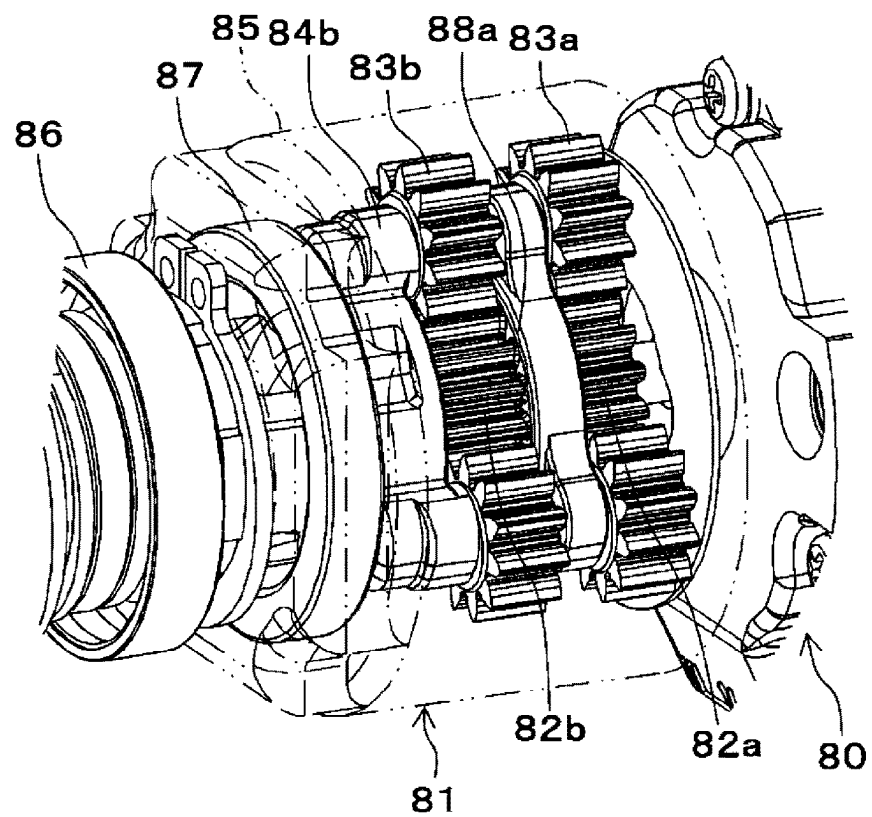


FIG. 6A

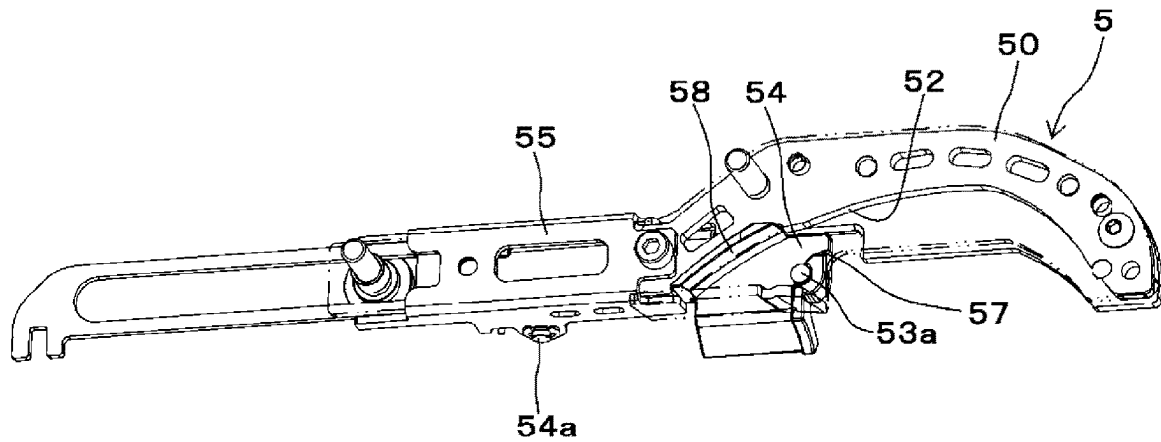


FIG. 6B

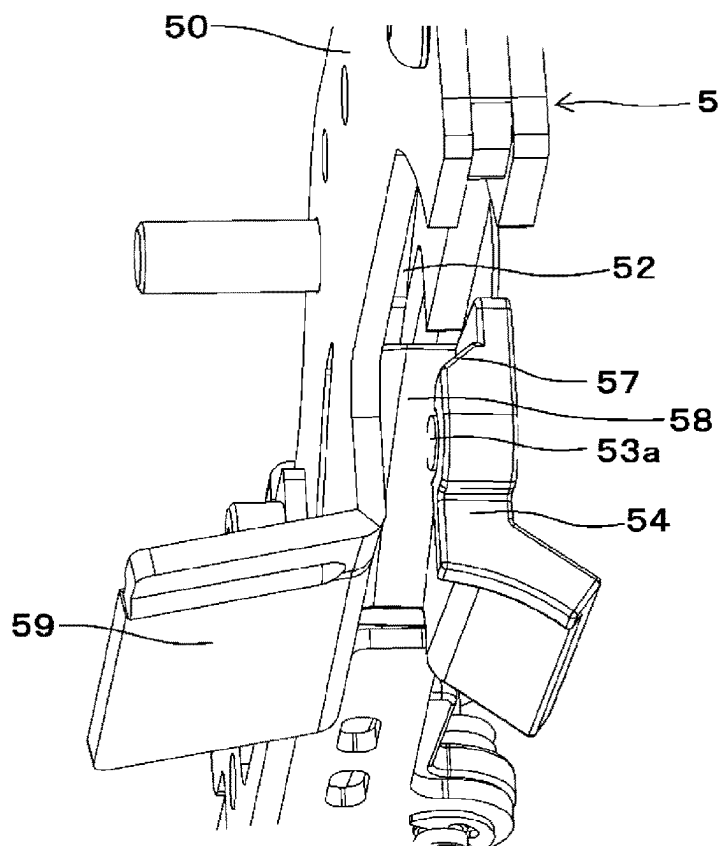


FIG. 6C

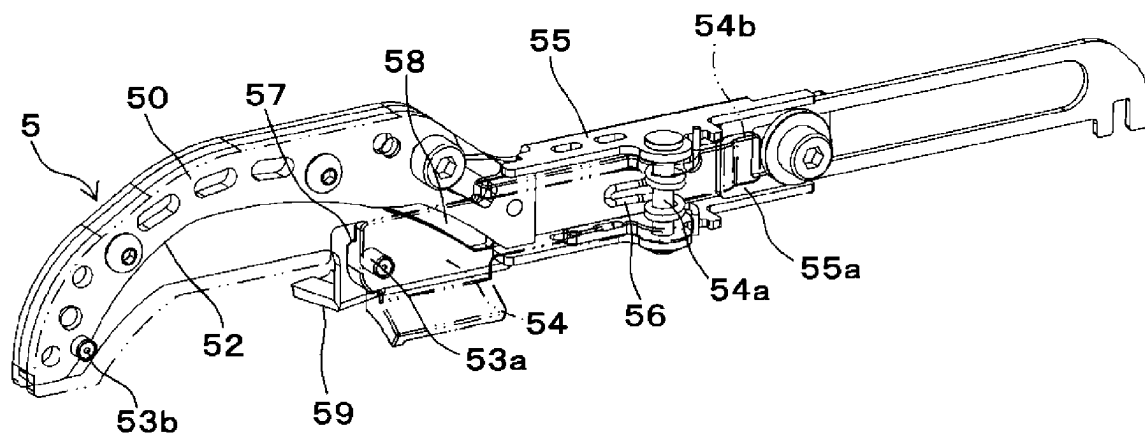


FIG. 6D

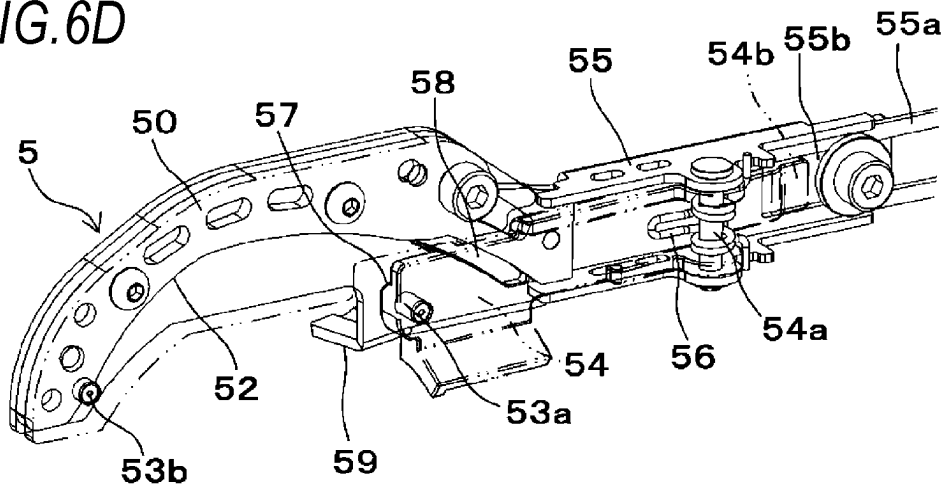


FIG. 7A

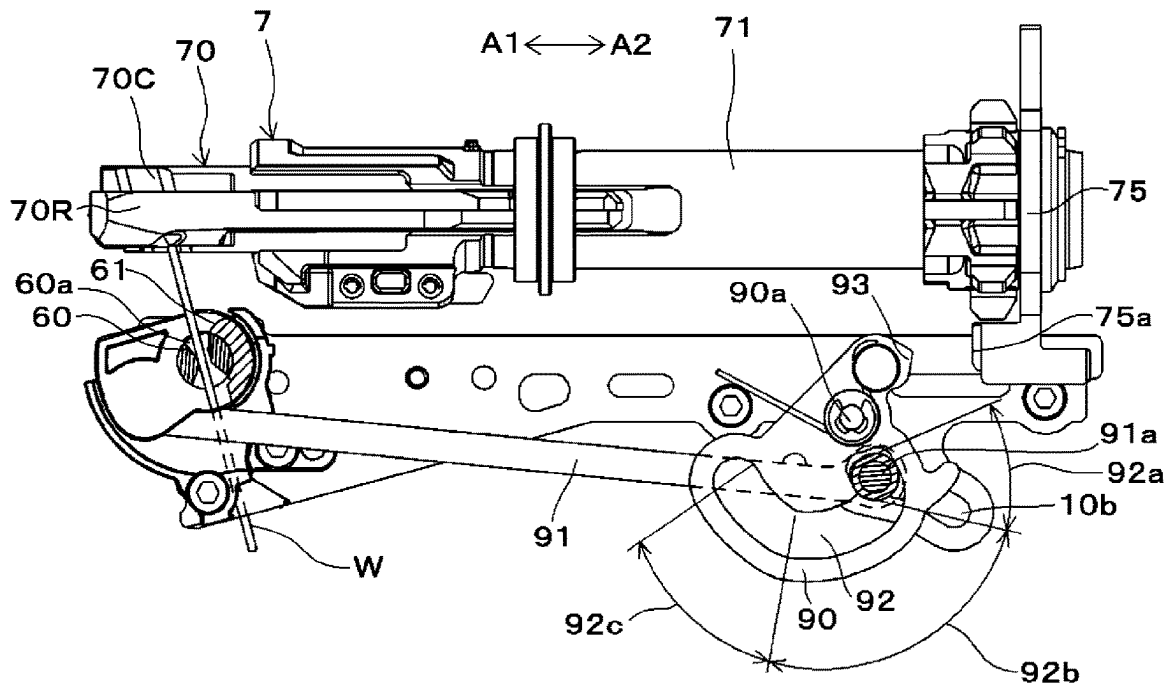


FIG. 7B

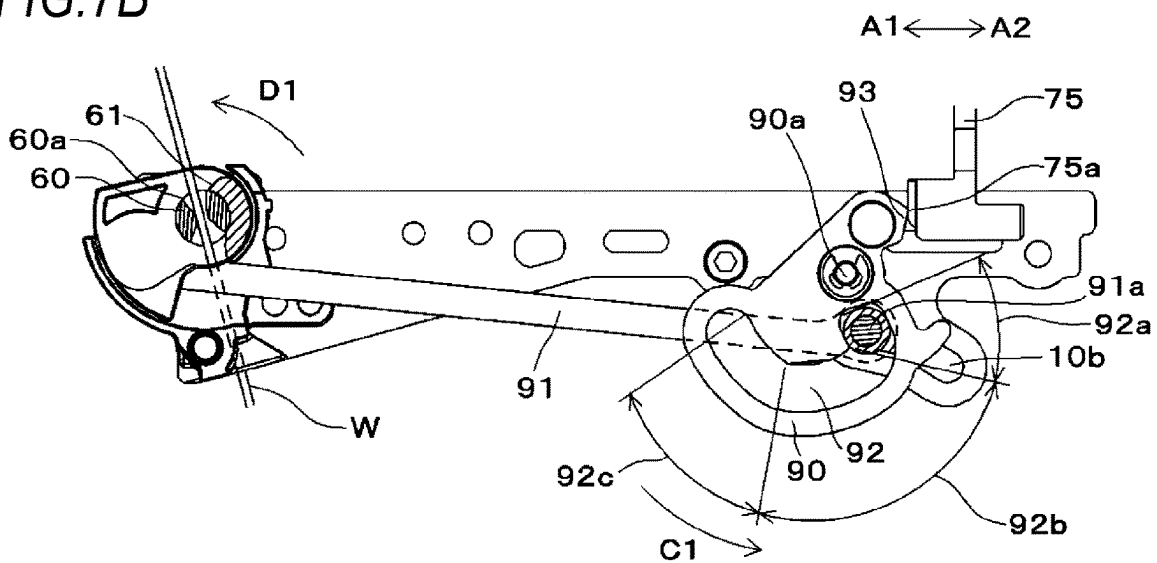


FIG. 7C

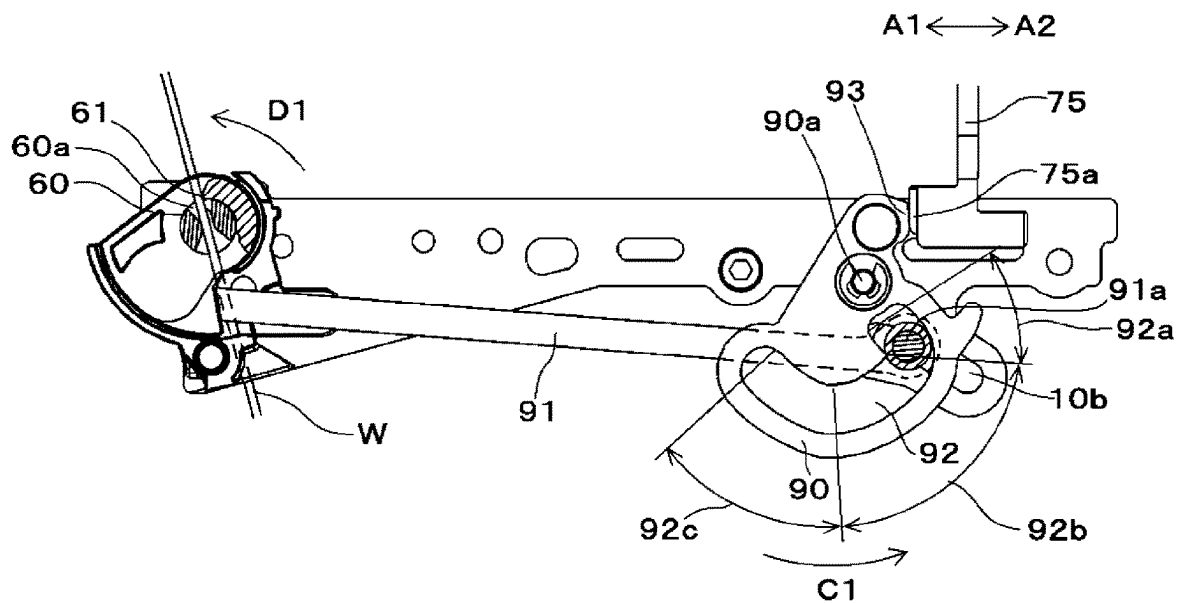


FIG. 7D

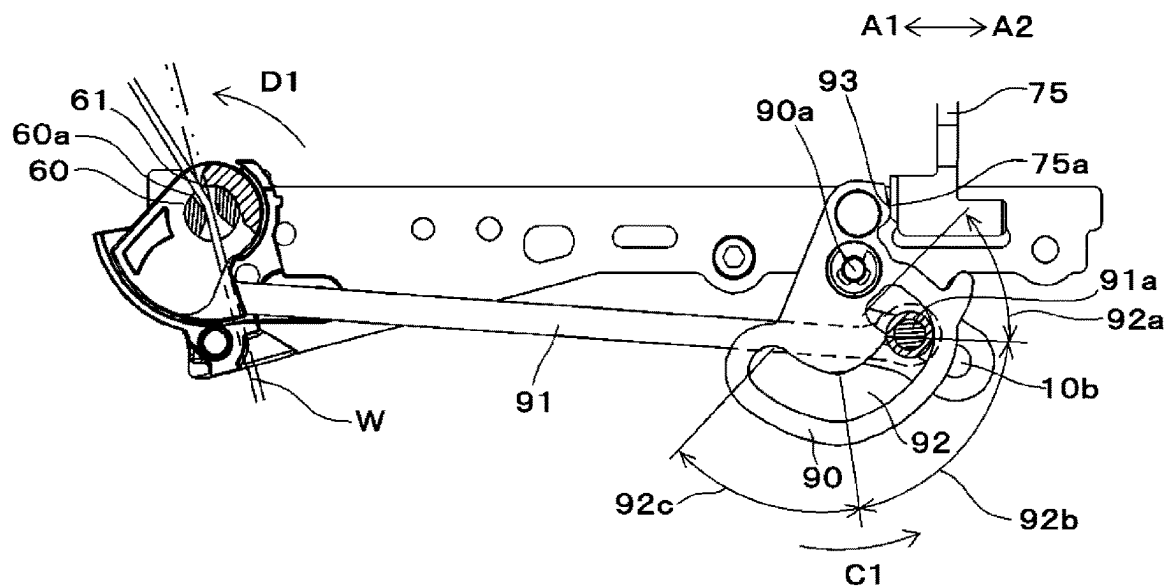


FIG. 7E

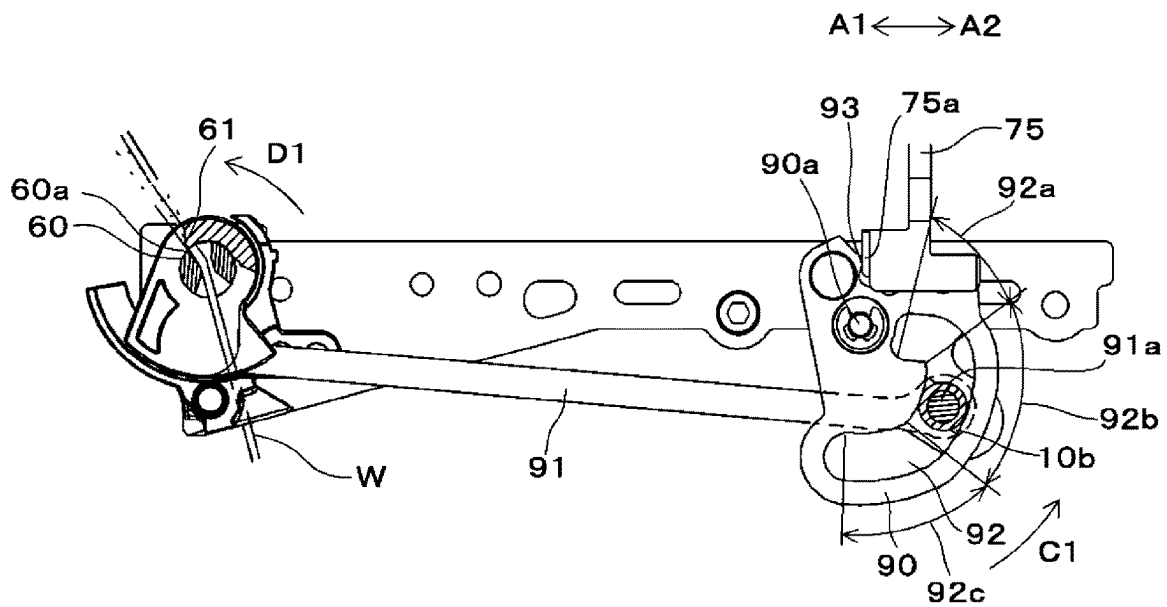


FIG. 7F

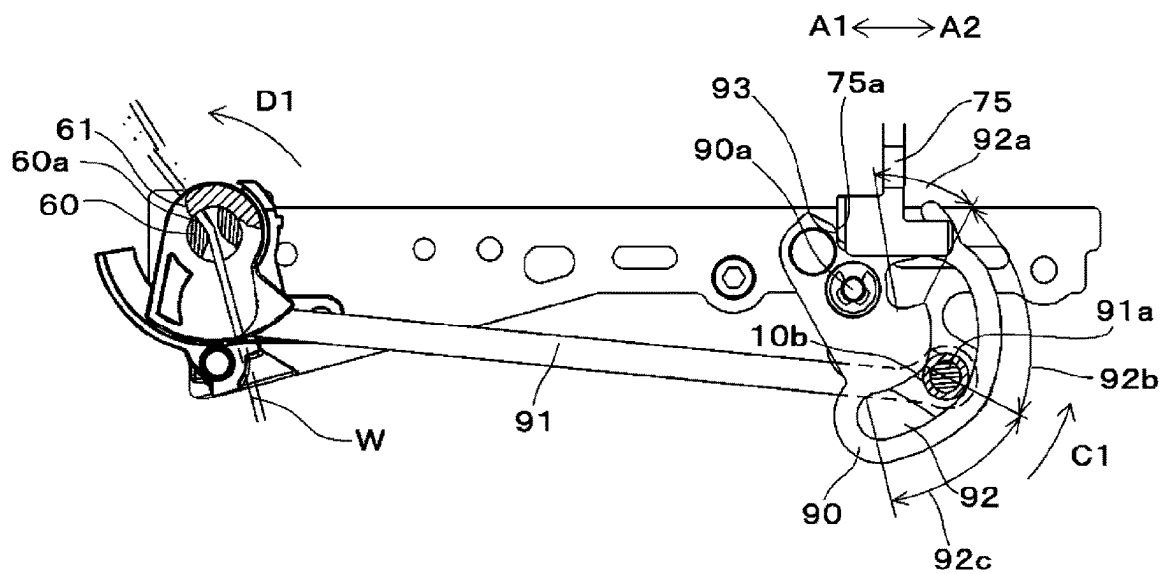


FIG. 7G

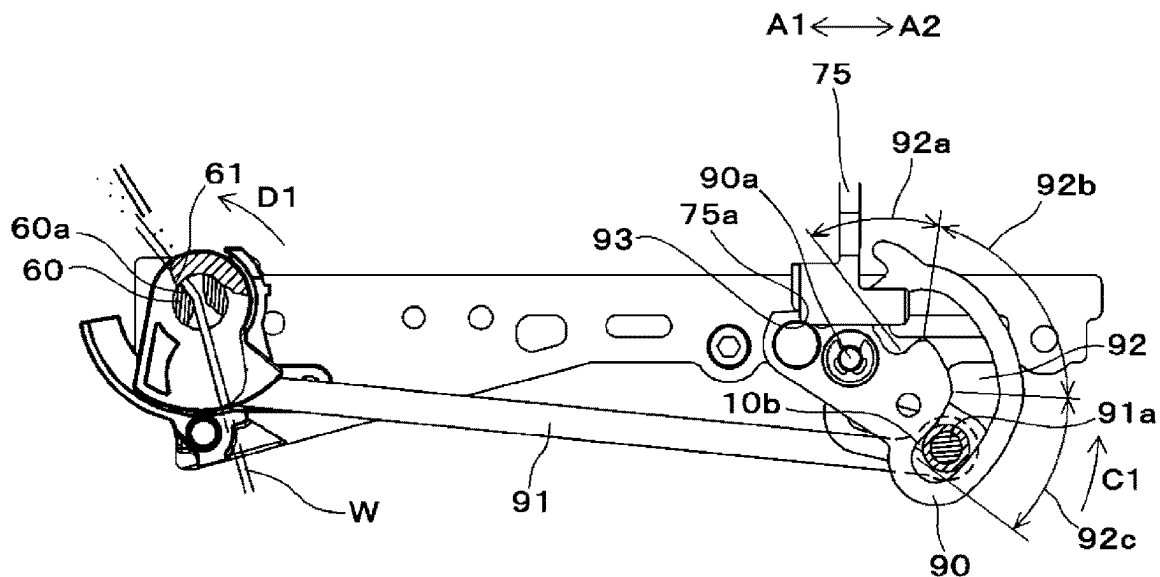


FIG. 8A

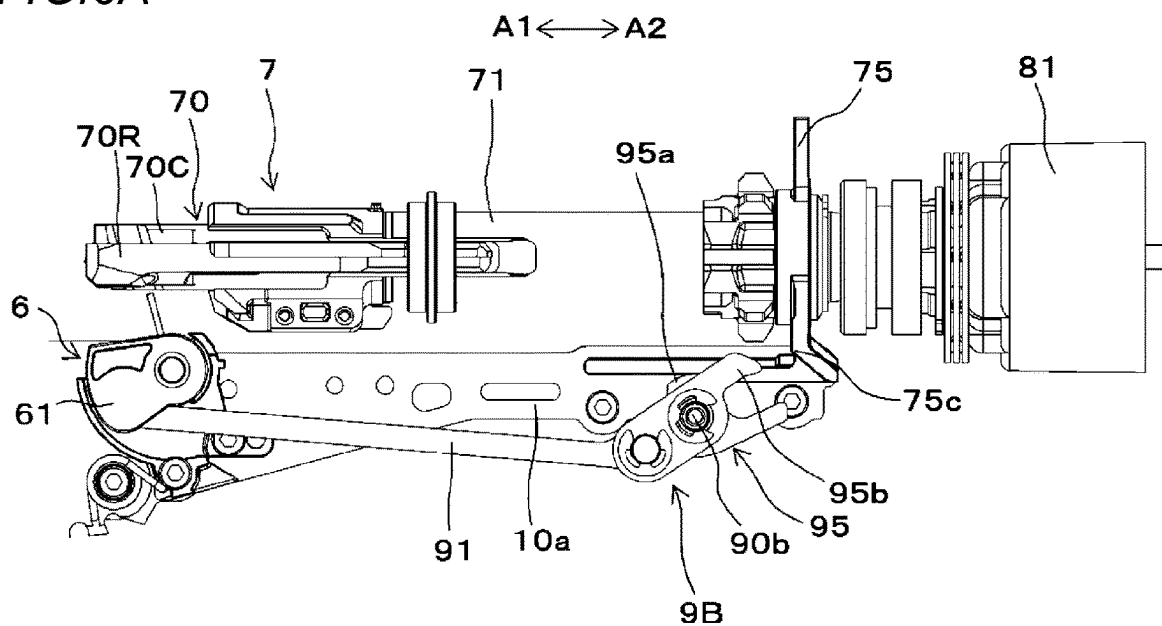


FIG. 8B

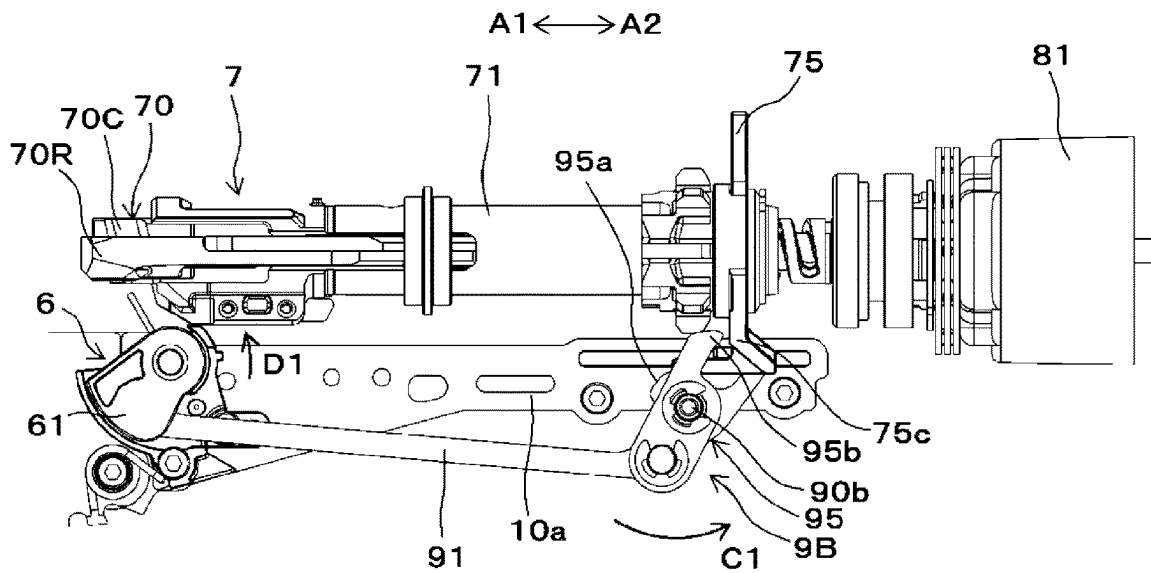


FIG. 8C

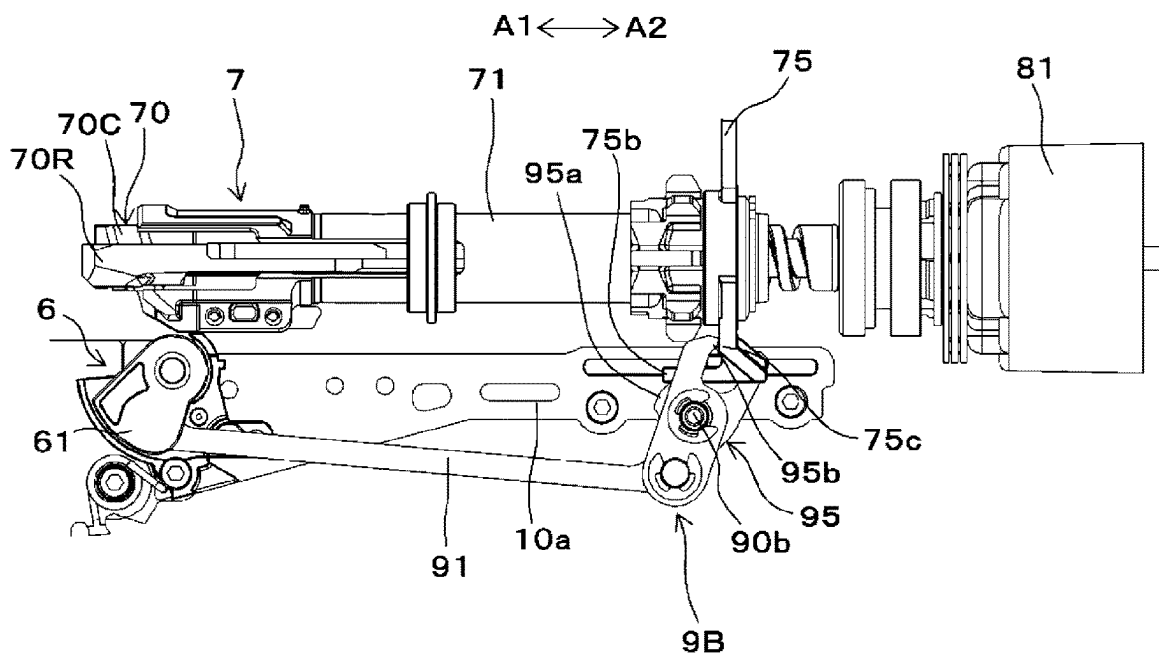


FIG. 9A

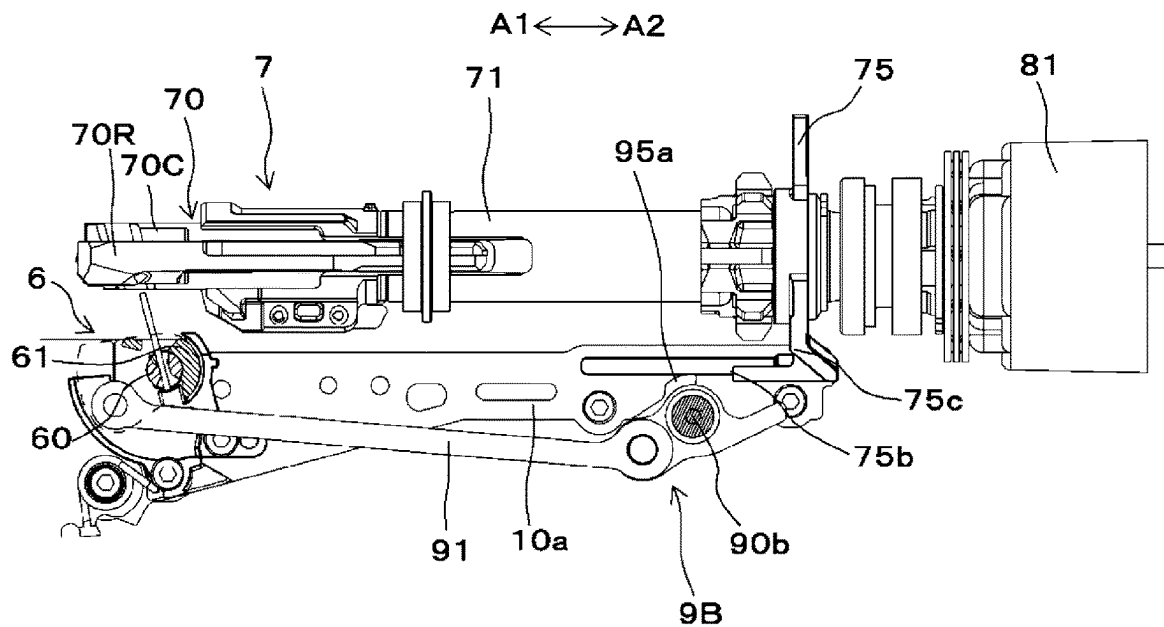
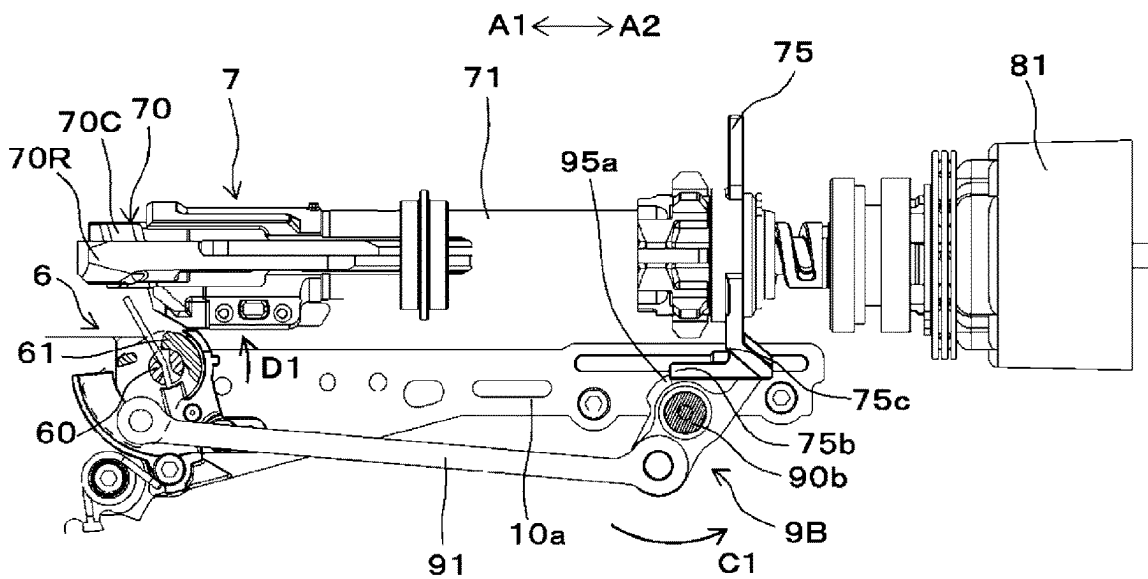


FIG. 9B



1

BINDING MACHINE

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application claims priority from Japanese Patent Application No. 2021-171966 filed on Oct. 20, 2021, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

The present invention relates to a binding machine configured to bind a to-be-bound object such as a reinforcing bar with a wire.

BACKGROUND ART

For concrete buildings, reinforcing bars are used so as to improve strength. The reinforcing bars are bound with wires so that the reinforcing bars do not deviate from predetermined positions during concrete placement.

In the related art, suggested is a binding machine referred to as a reinforcing bar binding machine configured to wind a wire on two or more reinforcing bars and to twist the wire wound on the reinforcing bars, thereby binding the two or more reinforcing bars with the wire.

The reinforcing bar binding machine includes a wire feeding unit configured to feed a wire, a curl forming unit configured to form a path along which the wire is to be wound around a to-be-bound object, a cutting unit configured to cut the wire, a binding unit configured to twist the wire, and a drive unit configured to drive the binding unit, in which the drive unit includes a motor and a decelerator configured to perform deceleration and amplification of torque. The decelerator is configured by a planet gear in which an input shaft and an output shaft are coaxially arrayed, and includes a sun gear attached to the input shaft, a planetary gear in mesh with the sun gear, a planet cage configured to support the planetary gear, and an internal gear in mesh with the planetary gear (for example, refer to JP2018-108850A).

As for the decelerator used in the binding machine, one side of the planet cage along an axis direction is rotatably supported by a main body part to which the internal gear is fixed. In contrast, the other side of the planet cage along the axis direction is in mesh with the planetary gear and the internal gear.

Since a predetermined clearance is required for the mesh between gears, there is a clearance in the mesh between the planetary gear and the internal gear, and there is a possibility that the planet cage will be inclined with respect to the axis direction.

The present invention has been made so as to solve the problem, and an object thereof is to provide a binding machine capable of suppressing a planet cage configured to support a planetary gear in a planet gear from being inclined.

SUMMARY

According to an aspect of the present invention, there is provided a binding machine including a wire feeding unit configured to feed a wire, a curl forming unit configured to form a path along which the wire fed by the wire feeding unit is to be wound around an object, a cutting unit configured to cut the wire wound on the object, a binding unit configured to twist the wire wound on the object and cut by the cutting

2

unit, and a drive unit configured to drive the binding unit. The drive unit comprises a motor and a decelerator configured to perform deceleration and amplification of torque. The decelerator includes a sun gear, a planetary gear in mesh with the sun gear, a planet cage configured to support the planetary gear, and an internal gear in mesh with the planetary gear. The planet cage is rotatably supported on one side and the other side along an axis direction.

According to another aspect of the present invention, there is provided a binding machine including a wire feeding unit configured to feed a wire, a curl forming unit configured to form a path along which the wire fed by the wire feeding unit is to be wound around an object, a cutting unit configured to cut the wire wound on the object, a binding unit configured to twist the wire wound on the object and cut by the cutting unit, and a drive unit configured to drive the binding unit. The drive unit includes a motor and a decelerator configured to perform deceleration and amplification of torque. The decelerator includes a first sun gear attached to a shaft of the motor, a first planetary gear in mesh with the first sun gear, a first planet cage configured to support the first planetary gear, a second sun gear provided to the first planet cage, a second planetary gear in mesh with the second sun gear, and a second planet cage configured to support the second planetary gear. The second planet cage is rotatably supported on one side and the other side along an axis direction.

In the present invention, the decelerator is configured by the planet gear in which the input shaft and the output shaft are coaxially arrayed, and the planet cage configuring the output shaft is rotatably supported on one side and the other side along the axis direction.

According to the present invention, with a simple configuration, the planet cage is suppressed from being inclined with respect to the axis direction, and changes in meshes between the sun gear and the planetary gear and between the planetary gear and the internal gear, and interferences between gears aligned in the axis direction, between a gear and a planet cage, and the like can be suppressed.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is an internal configuration view showing an example of an overall configuration of a reinforcing bar binding machine of the present embodiment, as seen from a side.

FIG. 2A is an internal configuration view showing an example of a main part configuration of the reinforcing bar binding machine of the present embodiment, as seen from a side.

FIG. 2B is an internal configuration view showing an example of the main part configuration of the reinforcing bar binding machine of the present embodiment, as seen from a side.

FIG. 2C is an internal configuration view showing an example of the main part configuration of the reinforcing bar binding machine of the present embodiment, as seen from a side.

FIG. 3A is a plan view showing an example of a binding unit according to the present embodiment.

FIG. 3B is a plan view showing the example of the binding unit according to the present embodiment.

FIG. 3C is a plan view showing the example of the binding unit according to the present embodiment.

FIG. 3D is a plan view of main parts showing a modified embodiment of the binding unit according to the present embodiment.

3

FIG. 3E is a plan view of main parts showing a modified embodiment of the binding unit according to the present embodiment.

FIG. 3F is a plan view of main parts showing a modified embodiment of the binding unit according to the present embodiment.

FIG. 4A is a plan view showing an example of a cutting unit according to the present embodiment.

FIG. 4B is a plan view showing the example of the cutting unit according to the present embodiment.

FIG. 4C is a perspective view showing the example of the cutting unit of the present embodiment.

FIG. 4D is a perspective view showing the example of the cutting unit of the present embodiment.

FIG. 4E is a perspective view showing the example of the cutting unit of the present embodiment.

FIG. 4F is a plan view showing a modified embodiment of the cutting unit according to the present embodiment.

FIG. 4G is a plan view showing a modified embodiment of the cutting unit according to the present embodiment.

FIG. 5A is a side cross-sectional view showing an example of a decelerator according to the present embodiment.

FIG. 5B is a perspective view showing the example of the decelerator according to the present embodiment.

FIG. 5C is a side cross-sectional view of main parts showing a modified embodiment of the decelerator according to the present embodiment.

FIG. 5D is a perspective view showing the modified embodiment of the decelerator according to the present embodiment.

FIG. 6A is a plan view showing an example of a curl forming unit according to the present embodiment.

FIG. 6B is a plan view showing the example of the curl forming unit according to the present embodiment.

FIG. 6C is a plan view showing the example of the curl forming unit according to the present embodiment.

FIG. 6D is a plan view showing the example of the curl forming unit according to the present embodiment.

FIG. 7A is an operation explanatory diagram showing an example of operations of the binding unit, a transmission unit and the cutting unit according to the present embodiment.

FIG. 7B is an operation explanatory diagram showing the example of operations of the binding unit, the transmission unit and the cutting unit according to the present embodiment.

FIG. 7C is an operation explanatory diagram showing the example of operations of the binding unit, the transmission unit and the cutting unit according to the present embodiment.

FIG. 7D is an operation explanatory diagram showing the example of operations of the binding unit, the transmission unit and the cutting unit according to the present embodiment.

FIG. 7E is an operation explanatory diagram showing the example of operations of the binding unit, the transmission unit and the cutting unit according to the present embodiment.

FIG. 7F is an operation explanatory diagram showing the example of operations of the binding unit, the transmission unit and the cutting unit according to the present embodiment.

FIG. 7G is an operation explanatory diagram showing the example of operations of the binding unit, the transmission unit and the cutting unit according to the present embodiment.

4

FIG. 8A is a side view showing a modified embodiment of the transmission unit according to the present embodiment.

FIG. 8B is a side view showing the modified embodiment of the transmission unit according to the present embodiment.

FIG. 8C is a side view showing the modified embodiment of the transmission unit according to the present embodiment.

FIG. 9A is a side cross-sectional view showing the modified embodiment of the transmission unit according to the present embodiment.

FIG. 9B is a side cross-sectional view showing the modified embodiment of the transmission unit according to the present embodiment.

FIG. 9C is a side cross-sectional view of parts showing the modified embodiment of the transmission unit according to the present embodiment.

DESCRIPTION OF EMBODIMENTS

Hereinafter, an example of a reinforcing bar binding machine as an embodiment of the binding machine of the present invention will be described with reference to the drawings.

<Overall Configuration Example of Reinforcing Bar Binding Machine of Present Embodiment>

FIG. 1 is an internal configuration view showing an example of an overall configuration of a reinforcing bar binding machine of the present embodiment, as seen from a side.

A reinforcing bar binding machine 1A is configured to feed a wire W in a forward direction denoted with an arrow F, to wind the wire around reinforcing bars S, which are a to-be-bound object (an object), to feed the wire W wound around the reinforcing bars S in a reverse direction denoted with an arrow R, to wind the wire on the reinforcing bars S, to cut the wire, and to twist the wire W, thereby binding the reinforcing bars S with the wire W.

The reinforcing bar binding machine 1A includes a magazine 2 in which the wire W is accommodated, a wire feeding unit 3 configured to feed the wire W, and a wire guide 4 configured to guide the wire W, so as to implement the above-described functions. In addition, the reinforcing bar binding machine 1A includes a curl forming unit 5 configured to form a path along which the wire W fed by the wire feeding unit 3 is to be wound around the reinforcing bars S, and a cutting unit 6 configured to cut the wire W wound on the reinforcing bars S. Further, the reinforcing bar binding machine 1A includes a binding unit 7 configured to twist the wire W wound on the reinforcing bars S, a drive unit 8 configured to drive the binding unit 7, and a transmission unit 9 configured to transmit an operation of the binding unit 7 to the cutting unit 6.

Further, the reinforcing bar binding machine 1A has such a form that an operator grips and uses with a hand, and has a main body part 10 and a handle part 11.

The magazine 2 is an example of the accommodation unit, and a reel 20 on which the long wire W is wound to be reeled out is rotatably and detachably accommodated therein. For the wire W, a wire made of a plastically deformable metal wire, a wire having a metal wire covered with a resin, or a twisted wire is used.

In a configuration in which the reinforcing bars S are bound with one wire W, one wire W is wound on a hub part (not shown) of the reel 20, and one wire W can be pulled out while the reel 20 rotates. In addition, in a configuration in

5

which the reinforcing bars S are bound with a plurality of wires W, the plurality of wires W are wound on the hub part, and the plurality of wires W can be pulled out at the same time while the reel 20 rotates. For example, in a configuration in which the reinforcing bars S are bound with two wires W, the two wires W are wound on the hub part, and the two wires W can be pulled out at the same time while the reel 20 rotates.

The wire feeding unit 3 includes a pair of feeding gears 30 configured to sandwich and feed the wire W. The wire feeding unit 3 is configured such that a rotating operation of a feeding motor (not shown) is transmitted to rotate the feeding gears 30. Thereby, the wire feeding unit 3 is configured to feed the wire W sandwiched between the pair of feeding gears 30 along an extension direction of the wire W. In a configuration in which a plurality of, for example, two wires W are fed to bind the reinforcing bars S, the two wires W are fed aligned in parallel.

The wire feeding unit 3 is configured such that a rotation direction of the feeding motor (not shown) is switched between forward and reverse directions to switch rotation directions of the feeding gears 30, thereby feeding the wire W in the forward direction denoted with the arrow F, feeding the wire W in the reverse direction denoted with the arrow R, or switching the feeding direction of the wire W between the forward and reverse directions.

The wire guide 4 is provided at a predetermined position on an upstream side and a downstream side of the wire feeding unit 3 with respect to a feeding direction of feeding the wire W in the forward direction, respectively. In the configuration in which the two wires W are fed to bind the reinforcing bars S, the wire guide 4 provided on the upstream side of the wire feeding unit 3 is configured to regulate the two wires W in a radial direction, to align the two introduced wires W in parallel and to guide the wires between the pair of feeding gears 30. The wire guide 4 provided on the downstream side of the wire feeding unit 3 is configured to regulate the two wires W in the radial direction, to align the two introduced wires W in parallel, and to guide the wires toward the cutting unit 6 and the curl forming unit 5.

The curl forming unit 5 includes a curl guide 50 configured to curl the wire W that is fed by the wire feeding unit 3, and an induction guide 51 configured to guide the wire W curled by the curl guide 50 toward the binding unit 7. In the reinforcing bar binding machine 1A, the path of the wire W that is fed by the wire feeding unit 3 is regulated by the curl forming unit 5, so that a locus of the wire W becomes a loop Ru as shown with a dashed-two dotted line in FIG. 1 and the wire W is thus wound around the reinforcing bars S.

In the reinforcing bar binding machine 1A, the curl guide 50 and the induction guide 51 of curl forming unit 5 are provided at an end portion on a front side of the main body part 10.

The cutting unit 6 includes a fixed blade part 60 and a movable blade part 61 configured to cut the wire W in cooperation with the fixed blade part 60. The cutting unit 6A is configured to cut the wire W by a rotating operation of the movable blade part 61 about the fixed blade part 60 as a fulcrum shaft. In the present specification, the cutting unit 6 is described as the fixed blade part 60 and the movable blade part 61 configured to rotate about the fixed blade part 60 as a fulcrum shaft. However, the movable blade part 61 may be of a slide type configured to linearly slide, not to rotate.

The transmission unit 9 includes a cam 90 configured to rotate by an operation of the binding unit 7, and a link 91 configured to connect the cam 90 and the movable blade part

6

61. The transmission unit 9 is configured to transmit the operation of the binding unit 7 to the movable blade part 61 of the cutting unit 6 via the cam 90 and the link 91.

The binding unit 7 includes a locking member 70 configured to lock the wire W, and a sleeve 71 configured to actuate the locking member 70. The drive unit 8 includes a motor 80, and a decelerator 81 configured to perform deceleration and amplification of torque.

The binding unit 7 is configured to be driven by the drive unit 8, whereby the sleeve 71 actuates the locking member 70 to lock the wire W. In addition, the binding unit 7 is configured to bind the reinforcing bars S by twisting the wire W after cutting the wire W by the cutting unit 6 in conjunction with the operation of the sleeve 71.

In the reinforcing bar binding machine 1A, the wire feeding unit 3, the wire guide 4, the cutting unit 6, the binding unit 7, the drive unit 8, the transmission unit 9, and the like are accommodated within the main body part 10. In the reinforcing bar binding machine 1A, the binding unit 7 is provided inside a front side of the main body part 10, and the drive unit 8 is provided inside a rear side. In addition, in the reinforcing bar binding machine 1A, a butting portion 16 against which the reinforcing bars S are to be butted is provided at an end portion on the front side of the main body part 10 and between the curl guide 50 and the induction guide 51.

Further, in the reinforcing bar binding machine 1A, the handle part 11 extends downward from the main body part 10, and a battery 15 is detachably mounted to a lower part of the handle part 11. In addition, in the reinforcing bar binding machine 1A, the magazine 2 is provided in front of the handle part 11.

In the reinforcing bar binding machine 1A, a trigger 12 is provided on a front side of the handle part 11, and a switch 13 is provided inside the handle part 11. In the reinforcing bar binding machine 1A, a control unit 14 is configured to control the motor 80 and a feeding motor (not shown), in response to a state of the switch 13 that is pressed by an operation on the trigger 12.

<Configuration Example of Main Parts of Reinforcing Bar Binding Machine of Present Embodiment>

FIGS. 2A to 2C are internal configuration views showing an example of a main part configuration of the reinforcing bar binding machine of the present embodiment, as seen from a side, in which FIG. 2A mainly shows the binding unit 7, the cutting unit 6 and the transmission unit 9, FIG. 2B is a cross-sectional view of the cutting unit 6 and the transmission unit 9 in FIG. 2A, and FIG. 2C shows the internal configuration by showing an outer shape of the sleeve 71 in FIG. 2A with a dashed-two dotted line. In addition, FIGS. 3A to 3C are plan views showing an example of the binding unit of the present embodiment, and FIGS. 3D to 3F are plan views of main parts showing modified embodiments of the binding unit of the present embodiment.

Example of Embodiment of Binding Unit

Next, an example of the binding unit of the present embodiment will be described with reference to each drawing. The binding unit 7 has a rotary shaft 72 configured to move and rotate the sleeve 71, thereby actuating the locking member 70. The binding unit 7 and the drive unit 8 are configured such that the rotary shaft 72 and the motor 80 are connected via the decelerator 81 and the rotary shaft 72 is driven by the motor 80 via the decelerator 81.

The locking member 70 includes a center hook 70C connected to the rotary shaft 72, and a first side hook 70R and a second side hook 70L configured to open/close with respect to the center hook 70C.

In the binding unit 7, a side on which the center hook 70C, the first side hook 70R and the second side hook 70L are provided is referred to as a front side, and a side on which the rotary shaft 72 is connected to the decelerator 81 is referred to as a rear side.

The center hook 70C is connected to a front end of the rotary shaft 72, which is one end portion, via a configuration that can rotate with respect to the rotary shaft 72, can rotate integrally with the rotary shaft 72 and can move integrally with the rotary shaft 72 in an axis direction.

A tip end side of the first side hook 70R, which is one end portion along the axis direction of the rotary shaft 72, is located on one side part with respect to the center hook 70C. In addition, a rear end side of the first side hook 70R, which is the other end portion along the axis direction of the rotary shaft 72, is rotatably supported to the center hook 70C by a shaft 71b.

A tip end side of the second side hook 70L, which is one end portion along the axis direction of the rotary shaft 72, is located on the other side part with respect to the center hook 70C. In addition, a rear end side of the second side hook 70L, which is the other end portion along the axis direction of the rotary shaft 72, is rotatably supported to the center hook 70C by the shaft 71b.

Thereby, the locking member 70 is configured to open/close in directions in which the tip end side of the first side hook 70R is contacted/separated with respect to the center hook 70C by a rotating operation about the shaft 71b as a fulcrum. The locking member is also configured to open/close in directions in which the tip end side of the second side hook 70L is contacted/separated with respect to the center hook 70C.

The rotary shaft 72 is connected at a rear end, which is the other end portion, to the decelerator 81 via a connection portion 72b having a configuration of enabling the rotary shaft 72 to rotate integrally with the decelerator 81 and to move in the axis direction with respect to the decelerator 81. The connection portion 72b has a spring 72c for urging backward the rotary shaft 72 toward the decelerator 81, and regulating a position of the rotary shaft 72 along the axis direction. Thereby, the rotary shaft 72 is configured to be movable forward away from the decelerator 81 while receiving a force pushed backward by the spring 72c. Accordingly, the rotary shaft 72 and the locking member 70 connected to the rotary shaft 72 can move forward up to a predetermined amount defined by the connection portion 72b while receiving the force pushed backward by the spring 72c.

The sleeve 71 has such a shape that a range of a predetermined length along the axis direction of the rotary shaft 72 from an end portion in the forward direction denoted with the arrow A1 is divided into two in a radial direction and the first side hook 70R and the second side hook 70L enter. In addition, the sleeve 71 is formed in a cylindrical shape configured to cover around the rotary shaft 72, and has a convex portion (not shown) protruding from an inner peripheral surface of a cylinder-shaped space in which the rotary shaft 72 is inserted, and the convex portion enters a groove portion of a feeding screw 72a formed along the axis direction on an outer periphery of the rotary shaft 72.

When the rotary shaft 72 rotates, the sleeve 71 is moved in a front and rear direction along the axis direction of the rotary shaft 72 according to a rotation direction of the rotary shaft 72 by an action of the convex portion (not shown) and the feeding screw 72a of the rotary shaft 72. In addition, when the sleeve 71 is moved to a forward end portion of the feeding screw 72a along the axis direction of the rotary shaft 72, the sleeve is rotated integrally with the rotary shaft 72.

The sleeve 71 has an opening/closing pin 71a configured to open/close the first side hook 70R and the second side hook 70L. The first side hook 70R has an opening/closing guide hole 73R into which the opening/closing pin 71a is inserted, and the second side hook 70L has an opening/closing guide hole 73L into which the opening/closing pin 71a is inserted.

The opening/closing guide holes 73R and 73L are configured by grooves extending along a moving direction of the sleeve 71. The opening/closing guide hole 73R has an opening/closing portion 73a having a shape of converting linear motion of the opening/closing pin 71a configured to move in conjunction with the sleeve 71 into an opening/closing operation by rotation of the first side hook 70R about the shaft 71b as a fulcrum. In addition, the opening/closing guide hole 73L has an opening/closing portion 73a having a shape of converting linear motion of the opening/closing pin 71a configured to move in conjunction with the sleeve 71 into an opening/closing operation by rotation of the second side hook 70L about the shaft 71b as a fulcrum. The opening/closing portion 73a is configured by a groove inclined with respect to the moving direction of the sleeve 71 and the opening/closing pin 71a.

When the sleeve 71 is moved forward (denoted with the arrow A1) in a state where the first side hook 70R is opened with respect to the center hook 70C, the first side hook 70R is pushed by the opening/closing pin 71a, on an inner wall surface of the opening/closing portion 73a formed in the opening/closing guide hole 73R with respect to a direction in which the first side hook 70R is closed. Thereby, the first side hook 70R is rotated about the shaft 71b as a fulcrum and is moved toward the center hook 70C as denoted with the arrow H1.

When the sleeve 71 is moved backward (denoted with the arrow A2) in a state where the first side hook 70R is closed with respect to the center hook 70C, the first side hook 70R is pushed by the opening/closing pin 71a, on an outer wall surface of the opening/closing portion 73a formed in the opening/closing guide hole 73R with respect to a direction in which the first side hook 70R is opened. Thereby, the first side hook 70R is rotated about the shaft 71b as a fulcrum and is moved away from the center hook 70C as denoted with the arrow H2.

When the sleeve 71 is moved forward (denoted with the arrow A1) in a state where the second side hook 70L is opened with respect to the center hook 70C, the second side hook 70L is pushed by the opening/closing pin 71a, on an inner wall surface of the opening/closing portion 73a formed in the opening/closing guide hole 73L with respect to a direction in which the second side hook 70L is closed. Thereby, the second side hook 70L is rotated about the shaft 71b as a fulcrum and is moved toward the center hook 70C as denoted with the arrow H1.

When the sleeve 71 is moved backward (denoted with the arrow A2) in a state where the second side hook 70L is closed with respect to the center hook 70C, the second side hook 70L is pushed by the opening/closing pin 71a, on an outer wall surface of the opening/closing portion 73a formed in the opening/closing guide hole 73L with respect to a direction in which the second side hook 70L is opened. Thereby, the second side hook 70L is rotated about the shaft 71b as a fulcrum and is moved away from the center hook 70C as denoted with the arrow H2.

The opening/closing guide hole 73L provided in the second side hook 70L has a locking portion 73b and an unlocking portion 73c. The opening/closing guide hole 73L is formed with the locking portion 73b on a downstream side

of the opening/closing portion **73a** and is formed with the unlocking portion **73c** on a downstream side of the locking portion **73b**, with respect to the forward moving direction of the sleeve **71** denoted with the arrow **A1**.

The locking portion **73b** is formed on the inner wall surface of the opening/closing guide hole **73L** facing toward the direction of the arrow **H1**, which is the direction in which the second side hook **70L** is closed. The locking portion **73b** faces the outer wall surface of the opening/closing guide hole **73L** with a dimension substantially equivalent to a diameter of the opening/closing pin **71a**, and extends in parallel to the outer wall surface.

The unlocking portion **73c** is configured by providing the inner wall surface of the opening/closing guide hole **73L** with a concave portion that is concave with respect to the lock portion **73b**. The unlocking portion **73c** faces the outer wall surface of the opening/closing guide hole **73L** with a dimension slightly greater than the diameter of the opening/closing pin **71a**, and extends in parallel to the outer wall surface.

As shown in FIG. 3B, the second side hook **70L** is configured to lock the wire **W** in a state in which it does not allow movement of the wire **W** within a range in which the opening/closing pin **71a** is located at the locking portion **73b** of the opening/closing guide hole **73L**. Here, within the range in which the opening/closing pin **71a** is located at the locking portion **73b** of the opening/closing guide hole **73L**, operations of feeding the wire **W** in the reverse direction and winding the wire on the reinforcing bars **S** are performed, as described later.

On the other hand, within a range in which the opening/closing pin **71a** is moved in the direction of the arrow **A1** in conjunction with the sleeve **71** and the opening/closing pin **71a** is located at the unlocking portion **73c** of the opening/closing guide hole **73L**, as shown in FIG. 3C, the second side hook **70L** becomes movable in a direction of the arrow **H2** in which the second side hook **70L** is spaced apart from the center hook **70C** by such a predetermined amount that the wire **W** does not come off between the second side hook **70L** and the center hook **70C**.

The sleeve **71** has a bending portion **71c1** configured to form the wire **W** into a predetermined shape by pushing and bending a tip end side of the wire **W**, which is one end portion, in a predetermined direction. In addition, the sleeve **71** has a bending portion **71c2** configured to form the wire **W** into a predetermined shape by pushing and bending a terminal end side, which is the other end portion of the wire **W** cut by the cutting unit **6**, in a predetermined direction. The bending portion **71c1** and the bending portion **71c2** are formed at an end portion of the sleeve **71** in the forward direction denoted with the arrow **A1**.

The sleeve **71** is moved in the forward direction denoted with the arrow **A1**, so that the tip end side of the wire **W** locked by the center hook **70C** and the second side hook **70L** is pushed and bent toward the reinforcing bars **S** by the bending portion **71c1**. In addition, the sleeve **71** is moved in the forward direction denoted with the arrow **A1**, so that the terminal end side of the wire **W** locked by the center hook **70C** and the first side hook **70R** and cut by the cutting unit **6** is pushed and bent toward the reinforcing bars **S** by the bending portion **71c2**.

The binding unit **7** includes a rotation regulation part **74** configured to regulate rotations of the locking member **70** and the sleeve **71** in conjunction with the rotating operation of the rotary shaft **72**. The rotation regulation part **74** has a rotation regulation blade **74a** provided to the sleeve **71**, and

a rotation regulation claw (not shown) to which the rotation regulation blade **74a** is locked and which is provided to the main body part **10**.

The rotation regulation blade **74a** is configured by a plurality of convex portions protruding radially from an outer periphery of the sleeve **71** and provided with predetermined intervals in a circumferential direction of the sleeve **71**. The rotation regulation blade **74a** is fixed to the sleeve **71** and is configured to move and rotate integrally with the sleeve **71**.

In an operation area in which the wire **W** is locked by the locking member **70**, the wire **W** is wound on the reinforcing bars **S** and is cut and further the wire **W** is bent and shaped by the bending portions **71c1** and **71c2** of the sleeve **71**, the rotation regulation blade **74a** of the rotation regulation part **74** is locked. When the rotation regulation blade **74a** is locked, the rotation of the sleeve **71** in conjunction with the rotation of the rotary shaft **72** is regulated, so that the sleeve **71** is moved in the front and rear direction by the rotating operation of the rotary shaft **72**.

In addition, in an operation area in which the wire **W** locked by the locking member **70** is twisted, the rotation regulation blade **74a** of the rotation regulation part **74** is unlocked. When the rotation regulation blade **74a** is unlocked, the sleeve **71** is rotated in conjunction with the rotation of the rotary shaft **72**. The center hook **70C**, the first side hook **70R** and the second side hook **70L** of the locking member **70** locking the wire **W** are rotated in conjunction with the rotation of the sleeve **71**. In an operation region of the sleeve **71** and the locking member **70** along the axis direction of the rotary shaft **72**, an operation region in which the wire **W** is locked by the locking member **70** is referred to as a first operation area. In addition, an operation area in which the wire **W** locked by the locking member **70** is twisted is referred to as a second operation area.

The binding unit **7** includes a moving member **75** configured to actuate the transmission unit **9**. The moving member **75** is rotatably attached to the sleeve **71**, and is configured not to operate in conjunction with the rotation of the sleeve **71** and to be movable in the front and rear direction in conjunction with the sleeve **71**.

The moving member **75** has an engaging portion **75a** configured to engage with the cam **90** of the transmission unit **9**. The engaging portion **75a** is configured not to operate in conjunction with the rotation of the sleeve **71**, and to move in the front and rear direction in conjunction with the sleeve **71**.

Note that, as a modified embodiment of the opening/closing guide hole **73L** provided in the second side hook **70L**, in a modified embodiment shown in FIG. 3D, the opening/closing guide hole **73L** may be configured to have a first locking portion **73b**, an unlocking portion **73c**, and a second locking portion **73d**. The opening/closing guide hole **73L** is formed with the first locking portion **73b** on a downstream side of the opening/closing portion **73a**, the unlocking portion **73c** on a downstream side of the first locking portion **73b**, and the second locking portion **73d** on a downstream side of the unlocking portion **73c**, with respect to the forward moving direction of the sleeve **71** denoted with the arrow **A1**.

The first locking portion **73b** and the second locking portion **73d** are formed in the inner wall surface of the opening/closing guide hole **73L** facing toward the direction of the arrow **H1**, which is the direction in which the second side hook **70L** is closed. The first locking portion **73b** and the second locking portion **73d** are configured to face the outer wall surface of the opening/closing guide hole **73L** with a

11

dimension substantially equivalent to the diameter of the opening/closing pin 71a, and extend in parallel to the outer wall surface.

The unlocking portion 73c is configured by providing the inner wall surface of the opening/closing guide hole 73L with a concave portion that is concave with respect to the first locking portion 73b and the second locking portion 73b. The unlocking portion 73c is configured to face the outer wall surface of the opening/closing guide hole 73L with a dimension slightly greater than the diameter of the opening/closing pin 71a, and extends in parallel to the outer wall surface.

In the modified embodiment shown in FIG. 3D, the second side hook 70L is configured to enable the opening/closing pin 71a to move along the inner wall surface of the opening/closing guide hole 73L by an operation of the opening/closing pin 71a moving in the direction of the arrow A1, and to lock the wire W in a state in which the wire W is not allowed to move, within a range in which the opening/closing pin 71a is located at the first locking portion 73b of the opening/closing guide hole 73L, as shown with the solid line.

On the other hand, within a range in which the opening/closing pin 71a is moved in the direction of the arrow A1 and the opening/closing pin 71a is located at the unlocking portion 73c of the opening/closing guide hole 73L, as shown with the dashed-two dotted line, the opening/closing guide hole 73L can be displaced up to a position denoted with the dashed-two dotted line, with respect to the opening/closing pin 71a, and the second side hook 70L becomes movable in the direction of the arrow H2 in which the second side hook 70L is spaced apart from the center hook 70C by such a predetermined amount that the wire W does not come off between the second side hook 70L and the center hook 70C.

Further, within a range in which the opening/closing pin 71a is moved in the direction of the arrow A1 and the opening/closing pin 71a is located at the second locking portion 73d of the opening/closing guide hole 73L, as shown with the broken line, the wire W is locked in a state in which the wire W is not allowed to move. Here, within the range in which the opening/closing pin 71a is located at the second locking portion 73d of the opening/closing guide hole 73L, an operation of twisting the wire W with the binding unit 7 is performed, as described later.

In a modified embodiment shown in FIG. 3E, the opening/closing guide hole 73L has a first locking portion 73b, an unlocking portion 73c, and a second locking portion 73d. The unlocking portion 73c is configured to face, at a portion connected to the first lock portion 73b, the outer wall surface of the opening/closing guide hole 73L with a dimension slightly greater than the diameter of the opening/closing pin 71a. In addition, the unlocking portion 73c is configured by an inclined surface inclined with respect to the outer wall surface, and is connected to the second lock portion 73d.

In the modified embodiment shown in FIG. 3E, the second side hook 70L is configured to enable the opening/closing pin 71a to move along the inner wall surface of the opening/closing guide hole 73L by an operation of the opening/closing pin 71a moving in the direction of the arrow A1, and to lock the wire W in a state in which the wire W is not allowed to move, within a range in which the opening/closing pin 71a is located at the first locking portion 73b of the opening/closing guide hole 73L, as shown with the solid line.

On the other hand, within a range in which the opening/closing pin 71a is moved in the direction of the arrow A1 and the opening/closing pin 71a is located at the unlocking

12

portion 73c of the opening/closing guide hole 73L, as shown with the dashed-two dotted line, the opening/closing guide hole 73L can be displaced up to a position denoted with the dashed-two dotted line, with respect to the opening/closing pin 71a, and the second side hook 70L becomes movable in the direction of the arrow H2 in which the second side hook 70L is spaced apart from the center hook 70C by such a predetermined amount that the wire W does not come off between the second side hook 70L and the center hook 70C. In addition, within a range in which the opening/closing pin 71a is located at the unlocking portion 73c of the opening/closing guide hole 73L, as the opening/closing pin 71a comes closer to the second locking portion 73d, a movable amount in the direction in which the second side hook 70L is spaced apart from the center hook 70C becomes smaller.

Further, within a range in which the opening/closing pin 71a is moved in the direction of the arrow A1 and the opening/closing pin 71a is located at the second locking portion 73d of the opening/closing guide hole 73L, as shown with the broken line, the wire W is locked in a state in which the wire W is not allowed to move.

In a modified example shown in FIG. 3F, the opening/closing guide hole 73L has a first locking portion 73b, an unlocking portion 73c, and a second locking portion 73d. The unlocking portion 73c is configured to face, at a portion connected to the first lock portion 73b, the outer wall surface of the opening/closing guide hole 73L with a dimension slightly greater than the diameter of the opening/closing pin 71a. In addition, the unlocking portion 73c is configured by an inclined surface inclined with respect to the outer wall surface, and is connected to the second lock portion 73d.

The second locking portion 73d is configured by an inclined surface connected to the unlocking portion 73c. The second locking portion 73d is configured such that an interval between the inner wall surface and the outer wall surface of the opening/closing guide hole 73L becomes smaller toward the front side of the opening/closing guide hole 73L and the inner wall surface and the outer wall surface at an end portion on the front side of the opening/closing guide hole 73L face each other with a dimension substantially equivalent to the diameter of the opening/closing pin 71a.

In the modified embodiment shown in FIG. 3F, the second side hook 70L is configured to enable the opening/closing pin 71a to move along the inner wall surface of the opening/closing guide hole 73L by an operation of the opening/closing pin 71a moving in the direction of the arrow A1, and to lock the wire W in a state in which the wire W is not allowed to move, within a range in which the opening/closing pin 71a is located at the first locking portion 73b of the opening/closing guide hole 73L, as shown with the solid line.

On the other hand, within a range in which the opening/closing pin 71a is moved in the direction of the arrow A1 and the opening/closing pin 71a is located at the unlocking portion 73c of the opening/closing guide hole 73L, as shown with the dashed-two dotted line, the opening/closing guide hole 73L can be displaced up to a position denoted with the dashed-two dotted line, with respect to the opening/closing pin 71a, and the second side hook 70L becomes movable in the direction of the arrow H2 in which the second side hook 70L is spaced apart from the center hook 70C by such a predetermined amount that the wire W does not come off between the second side hook 70L and the center hook 70C. In addition, within a range in which the opening/closing pin 71a is located at the unlocking portion 73c of the opening/closing guide hole 73L, as the opening/closing pin 71a

13

comes closer to the second locking portion **73d**, a movable amount in the direction in which the second side hook **70L** is spaced apart from the center hook **70C** becomes smaller.

Further, within a range in which the opening/closing pin **71a** is moved in the direction of the arrow **A1** and the opening/closing pin **71a** is located at the second locking portion **73d** of the opening/closing guide hole **73L**, as shown with the broken line, the wire **W** is locked in a state in which the wire **W** is not allowed to move.

Example of Embodiment of Cutting Unit

FIGS. **4A** and **4B** are plan views showing an example of the cutting unit of the present embodiment, FIGS. **4C** to **4E** are perspective views showing the example of the cutting unit of the present embodiment, and FIGS. **4F** and **4G** are plan views showing modified embodiments of the cutting unit of the present embodiment. Next, an example of the cutting unit of the present embodiment will be described with reference to each drawing.

The fixed blade part **60** is an example of the blade part, has a cylindrical shape serving as an axis of rotation of the movable blade part **61**, and is provided with an opening **60a** penetrating in a radial direction of the cylindrical shape along the feeding path of the wire **W**. The opening **60a** has a shape through which the wire **W** can pass. In the configuration in which the reinforcing bars **S** are bound with the two wires **W**, a cross-sectional shape of the opening **60a** is a long hole shape along a direction in which the two wires **W** are aligned in parallel.

Preferably, the opening **60a** has, for example, a tapered shape in which opening areas on an introduction side and a discharge side of the opening **60a** are widened with respect to the feeding of the wire **W** in the forward direction denoted with the arrow **F**. The fixed blade part **60** is provided on a downstream side of the wire guide **4** with respect to the feeding direction of the wire **W** that is conveyed in the forward direction.

In the configuration in which the reinforcing bars **S** are bound with the two wires **W**, the fixed blade part **60** has a first butting portion **60b** and a second butting portion **60c** at an end portion of the opening **60a** exposed on a circumferential surface on which the movable blade part **61** slides. The fixed blade part **60** is provided with a plurality of butting portions in a direction in which a plurality of wires **W** are aligned in parallel, and in the present example, is provided with the first butting portion **60b**, which is one butting portion, and the second butting portion **60c**, which is the other butting portion, along the direction in which the two wires **W** are aligned in parallel.

The fixed blade part **60** is provided with the first butting portion **60b** on a front side and the second butting portion **60c** on an inner side, with respect to a moving direction of the movable blade part **61** denoted with an arrow **D1**. The fixed blade part **60** has a step portion **60d** formed between the first butting portion **60b** and the second butting portion **60c** by recessing the second butting portion **60c** with respect to the moving direction of the movable blade part **61** denoted with the arrow **D1**. A recessed amount is preferably about a half of the diameter of the wire **W**.

The fixed blade part **60** has a regulation portion **60e** configured to suppress the wire **W** butted against the first butting portion **60b** from moving in a direction of the second butting portion **60c**. The regulation portion **60e** is a planar surface extending in a direction substantially orthogonal to the moving direction of the movable blade part **61** denoted with the arrow **D1**, and is provided between the first butting portion **60b** and the step portion **60d**.

14

The movable blade part **61** is an example of the blade part, has a shape of sliding along the circumferential surface of the fixed blade part **60**, and is configured to be in sliding contact with an open end of the opening **60a** of the fixed blade part **60** by a rotating operation about the fixed blade part **60** serving as a fulcrum shaft.

The cutting unit **6** has wall portions **62a** and **62b** configured to regulate introduction of foreign matters. The wall portions **62a** and **62b** are provided on upstream and downstream sides along a locus of the rotating operation of the movable blade part **61**, with respect to the opening **60a** of the fixed blade part **60**. The wall portions **62a** and **62b** each have a shape following the locus of the rotating operation of the movable blade part **61** about the fixed blade part **60** serving as a fulcrum, and are configured to suppress foreign matters, such as wastes entering from an opening at a front end of the main body part **10** and shavings resulting from rubbing of the wire **W** and the reinforcing bar **S**, from entering the periphery of the movable blade part **61**. Thereby, it is possible to suppress a malfunction of the movable blade part **61** and an increase in load for rotating the movable blade part **61**.

As for the cutting unit **6**, when the movable blade part **61** is rotated in the direction of the arrow **D1** from an initial position, the wire **W** having passed through the opening **60a** of the fixed blade part **60** is pressed against the open end of the opening **60a** by the movable blade part **61**. One wire **W** of the two wires **W** aligned in parallel is pressed against an end edge portion of the first butting portion **60b** of the fixed blade part **60** by the operation of the movable blade part **61**, and the other wire **W** is introduced into the second butting portion **60c** of the fixed blade part **60**. Thereby, a shearing force is applied to one wire **W**, and cutting of the one wire **W** is started prior to the other wire **W**.

When the movable blade part **61** is rotated in the direction of the arrow **D1** to start cutting of the first wire **W**, which is one wire, and the first wire **W** is cut to a predetermined position, the second wire **W**, which is the other wire, is pressed against an end edge portion of the second butting portion **60c** of the fixed blade part **60** by the operation of the movable blade part **61**.

Thereby, cutting of the second wire **W** is started. Preferably, the shapes and positions of the first butting portion **60b** and the second butting portion **60c** are set so that, after starting the cutting of the first wire **W**, when the first wire **W** is cut in half or more in the radial direction, cutting of the second wire **W** is started. That is, a distance from the end edge portion of the first butting portion **60b** to the end edge portion of the second butting portion **60c** along the rotation direction of the movable blade part **61** denoted with the arrow **D1** is set to be a substantial half of the wire **W** in the radial direction.

When the movable blade part **61** is further rotated in the direction of the arrow **D1**, the cutting of the one wire **W** for which cutting has been started first is completed. When the movable blade part **61** is further rotated to a cutting completion position in the direction of arrow **D1**, the cutting of the other wire **W** for which cutting has been started later is completed.

The fixed blade part **60** has the regulation portion **60e** formed between the first butting portion **60b** and the second butting portion **60c** and having a planar surface extending in a direction substantially orthogonal to the moving direction of the movable blade part **61** denoted with the arrow **D1**. Due to the planar surface, when the movable blade part **61** is moved in the direction of the arrow **D1**, it is possible to

15

prevent an unintended force from acting on the wire W in the direction substantially orthogonal to the moving direction.

Thereby, the wire W butted against the first butting portion 60b by the movable blade part 61 is suppressed from moving to the direction of the second butting portion 60c. In addition, the wire W is suppressed from moving in the direction of the second butting portion 60c, so that wear of the step portion 60d is suppressed and a difference in distance from the end edge portion of the first butting portion 60b to the end edge portion of the second butting portion 60c along the rotation direction of the movable blade part 61 denoted with the arrow D1 is suppressed from decreasing. Therefore, it is possible to secure a phase difference of timings at which the cuttings of the two wires W are started, and to suppress an increase in load, which is caused when the cuttings of the two wires W are started at substantially the same time.

Note that, the regulation portion 60e may be configured by providing the planar surface, which extends in the direction substantially orthogonal to the moving direction of the movable blade part 61 denoted with the arrow D1, at a part between the first butting portion 60b and the step portion 60d. In addition, the regulation portion 60e may be configured by an inclined surface or a curved surface where the step portion 60d protrudes from the first butting portion 60b toward the second butting portion 60c along a direction (arrow D2) opposite to the moving direction of the movable blade part 61 denoted with the arrow D1.

Further, as shown in FIG. 4F, the regulation portion 60e may be configured by a convex portion protruding from the first butting portion 60b and the second butting portion 60c along the direction (arrow D2) opposite to the moving direction of the movable blade part 61 denoted with the arrow D1, between the first butting portion 60b and the second butting portion 60c. Thereby, the first butting portion 60b becomes a concave shape, so that the wire W butted against the first butting portion 60b by the movable blade part 61 is suppressed from moving in the direction of the second butting portion 60c.

Further, as shown in FIG. 4G, the regulation portion 60e may be formed into a shape of partitioning the first butting portion 60b and the second butting portion 60c therebetween. Thereby, the first butting portion 60b and the second butting portion 60c are made independent, so that the wire W butted against the first butting portion 60b by the movable blade part 61 is suppressed from moving in the direction of the second butting portion 60c.

Example of Embodiment of Transmission Unit

Next, an example of the transmission unit 9 of the present embodiment will be described with reference to each drawing. The transmission unit 9 is supported so that the cam 90 can rotate about a shaft 90a as a fulcrum. The shaft 90a is attached to a frame 10a attached to an interior of the main body part 10. The frame 10a has a guide portion 10b configured to regulate a moving direction of a link 91. The guide portion 10b is configured by a long hole penetrating through the plate-shaped frame 10a.

The cam 90 is an example of the displacement member, and has a cam groove 92 whose length from the shaft 90a is displaced. The cam groove 92 extends in radial and circumferential directions of the cam 90 about the shaft 90a, and intersects the guide portion 10b of the frame 10a. The cam groove 92 penetrates through the plate-shaped cam 90, so that an intersection of the cam groove 90 and the guide portion 10b communicates.

The cam 90 is configured such that a rotating operation about the shaft 90a as a fulcrum changes a portion of the cam

16

groove 92 intersecting the guide portion 10b, thereby changing a length from the shaft 90a to the intersection of the cam groove 92 and the guide portion 10b.

For the cam 90, ranges in which an amount of change in length between the shaft 90a and the cam groove 92 by the rotating operation about the shaft 90a as a fulcrum is large and small for the same amount of rotation of the cam 90 are set. In the present example, a first range 92a in which the amount of change in length between the shaft 90a and the cam groove 92 is the largest, a second range 92b in which the amount of change in length between the shaft 90a and the cam groove 92 is smaller than the first range 92a, and a third range 92c in which there is little amount of change in length between the shaft 90a and the cam groove 92 are provided.

The cam 90 is configured such that, while the first range 92a of the cam groove 92 intersects the guide portion 10b by the rotating operation in the direction of the arrow C1 about the shaft 90a as a fulcrum, the length from the shaft 90a to the intersection of the cam groove 92 and the guide portion 10b is shorter and the amount of change in length between the shaft 90a and the cam groove 92 becomes larger, as compared with a case where the second range 92b intersects the guide portion 10b.

In addition, the cam 90 is configured such that, while the second range 92b of the cam groove 92 intersects the guide portion 10b by the rotating operation in the direction of the arrow C1 about the shaft 90a as a fulcrum, the length from the shaft 90a to the intersection of the cam groove 92 and the guide portion 10b is longer and the amount of change in length between the shaft 90a and the cam groove 92 becomes smaller, as compared with the case where the first range 92a intersects the guide portion 10b.

Further, the cam 90 is configured such that, while the third range 92c of the cam groove 92 intersects the guide portion 10b by the rotating operation in the direction of the arrow C1 about the shaft 90a as a fulcrum, the length from the shaft 90a to the intersection of the cam groove 92 and the guide portion 10b is substantially equivalent and the amount of change in length between the shaft 90a and the cam groove 92 is further smaller and substantially constant, as compared with the case where the second range 92b intersects the guide portion 10b.

The cam 90 has an engaged portion 93 to which movement of the sleeve 71 is transmitted via the moving member 75. The engaged portion 93 is provided on an opposite side to the cam groove 92 with the shaft 90a interposed therebetween, and is arranged on a locus of the engaging portion 75a by the movement of the moving member 75 in conjunction with the movement of the sleeve 71 in the front and rear direction denoted with the arrows A1 and A2. The engaged portion 93 is engaged with the engaging portion 75a of the moving member 75 by an operation in which the sleeve 71 is moved in the forward direction denoted with the arrow A1.

The cam 90 is urged by a spring 94 in the direction of the arrow C2 in which the first range 92a of the cam groove 92 intersects the guide portion 10b by the rotating operation about the shaft 90a as a fulcrum. The spring 94 is configured by, for example, a torsion coil spring attached to the shaft 90a. Note that, the rotation direction of the cam 90 denoted with the arrow C2 corresponds to a direction in which the movable blade part 61 connected by the link 91 returns from the cutting completion position to the initial position. In consideration of a case in which the cam 90 cannot rotate in the direction of the arrow C2 with the force of the spring 94 by the operation of the movable blade part 61 returning from

17

the cutting completion position to the initial position, the moving member 75 is provided with a pressing convex portion 76 and the cam 90 is provided with a pressed convex portion 96. When the moving member 75 is moved in the direction of the arrow A1 direction and the cam 90 is rotated until the movable blade part 61 is rotated to the cutting completion position, the pressing convex portion 76 and the pressed convex portion 96 face. By the operation of the sleeve 71 moving in the direction of the arrow A2, the pressing convex portion 76 pushes the pressed convex portion 96, so that the cam 90 can be forced to start rotating in the direction of the arrow C2.

The link 91 is an example of the transmission member, and has an end portion in the forward direction denoted with the arrow A1 connected to the movable blade part 61, and an end portion in the backward direction denoted with the arrow A2 connected to the cam 90.

The link 91 has a shaft portion 91a configured to enter the cam groove 92 of the cam 90 and the guide portion 10b of the frame 10a. The shaft portion 91a is configured by a rotary body 91a1 configured to enter the cam groove 92, and a shaft 91a2 configured to rotatably support the rotary body 91a1 and to be non-rotatable with respect to the link 91 that enters the guide portion 10b, and is inserted into the cam groove 92 and the guide portion 10b at the intersection of the cam groove 92 and the guide portion 10b. The shaft portion 91a is configured to move along the cam groove 92 and the guide portion 10b by the rotating operation of the cam 90 about the shaft 90a as a fulcrum. Here, by the rotating operation of the cam 90 about the shaft 90a as a fulcrum, a force that is applied in a circumferential direction of the rotary body 91a1 as the cam groove 92 and the rotary body 91a1 are slid and a force that is applied in a circumferential direction of the shaft 91a2 as the guide portion 10b and the shaft 91a2 are slid become forces in opposite directions. Therefore, in the shaft portion 91a, the rotary body 91a1 and the shaft 91a2 are configured as separate components. Note that, the shaft portion 91a may have a first rotary body configured to enter the cam groove 92, a second rotary body configured to enter the guide portion 10b, and a shaft configured to rotatably support the first rotary body and the second rotary body.

When the sleeve 71 is moved in the forward direction denoted with the arrow A1, the moving member 75 is moved in the forward direction denoted with the arrow A1 in conjunction with the sleeve 71. The moving member 75 is configured such that the engaging portion 75a is engaged with the engaged portion 93 of the cam 90 by the moving operation in the forward direction denoted with the arrow A1.

When the moving member 75 is further moved in the forward direction denoted with the arrow A1, the engaged portion 93 is pushed forward, so that the cam 90 is rotated in the direction of the arrow C1 about the shaft 90a as a fulcrum. When the cam 90 is rotated in the direction of the arrow C1, a portion of the cam groove 92 intersecting the guide portion 10b changes, and the length from the shaft 90a to the intersection of the cam groove 92 and the guide portion 10b changes in an increasing direction.

Thereby, when the cam 90 is rotated in the direction of the arrow C1 and the shaft portion 91a of the link 91 is moved along the cam groove 92 and the guide portion 10b, the shaft portion 91a is moved in a direction away from the shaft 90a of the cam 90.

The transmission unit 9 is configured such that, when the shaft portion 91a of the link 91 is moved in the direction

18

away from the shaft 90a of the cam 90, the rotating operation of the cam 90 is converted into movement along the extension direction of the link 91.

Thereby, the rotating operation of the cam 90 is transmitted to the movable blade part 61 via the link 91, so that the movable blade part 61 is rotated in the direction of the arrow D1. Therefore, the moving operation of the sleeve 71 in the forward direction rotates the movable blade part 61 in a predetermined direction to cut the wire W.

A period during which the first range 92a of the cam groove 92 intersects the guide portion 10b corresponds to a period after the movable blade part 61 of the cutting unit 6 starts rotation until the cutting of the first wire W is started. The period until the cutting of the first wire W is started corresponds to a region in which a load is low.

In addition, a period during which the second range 92b of the cam groove 92 intersects the guide portion 10b corresponds to a period after the movable blade part 61 of the cutting unit 6 rotates and the cutting of the first wire W is started until the cutting of the second wire W ends. The period after the cutting of the first wire W is started until the cutting of the second wire W ends corresponds to a region in which a load is high. Further, a period during which the third range 92c of the cam groove 92 intersects the guide portion 10b corresponds to a period during which the cutting of the second wire W ends and the rotation of the movable blade part 61 stops. In this way, with respect to the amount of movement of the moving member 75, it is not necessary to rotate the cutter having completed the wire cutting operation more than necessary.

Note that, in the above embodiment, the cam 90 has such a configuration that the length from the intersection of the cam groove 92, which is a first connection portion connected to the link 91, and the guide portion 10b to the shaft 90a is switched by the rotating operation about the shaft 90a as a fulcrum due to the shape of the cam groove 92.

Thereby, the cam 90 makes it possible to switch the amount of rotation (amount of movement) of the movable blade part 61 and the force that can be generated by the movable blade part 61, within the rotating range (moving range) of the movable blade part 61.

On the other hand, the cam 90 may be configured such that a length from the engaged portion 93, which is a second connection portion connected to the sleeve 71, to the shaft 90a is switched by the rotating operation about the shaft 90a as a fulcrum.

Example of Embodiment of Decelerator

FIG. 5A is a side cross-sectional view showing an example of the decelerator of the present embodiment, FIG. 5B is a perspective view showing the example of the decelerator of the present embodiment, FIG. 5C is a side cross-sectional view of main parts showing a modified embodiment of the decelerator of the present embodiment, and FIG. 5D is a perspective view showing the modified embodiment of the decelerator of the present embodiment. Next, an example of the decelerator of the present embodiment will be described with reference to each drawing.

The decelerator 81 is configured by a planet gear in which an input shaft and an output shaft are coaxially arrayed, and includes a first sun gear 82a attached to a shaft 80a of a motor 80 serving as an input shaft, a first planetary gear 83a in mesh with the first sun gear 82a and a first planet cage 84a configured to support the first planetary gear 83a.

In addition, the decelerator 81 includes a second sun gear 82b provided to the first planet cage 84a, a second planetary

gear **83b** in mesh with the second sun gear **82b**, and a second planet cage **84b** configured to support the second planetary gear **83b**.

Further, the decelerator **81** includes an internal gear **85** in mesh with the first planetary gear **83a** and the second planetary gear **83b**.

As for the decelerator **81**, the internal gear **85** is fixed to the main body part **10**. In addition, as for the decelerator **81**, the first planet cage **84a** and the second planet cage **84b** are arranged coaxially with the shaft **80a** of the motor **80**. Further, as for the decelerator **81**, the second planet cage **84b** is connected to the rotary shaft **72**, and configures an output shaft.

As for the decelerator **81**, a front side portion **84f** that is one side along an axis direction of the second planet cage **84b** protrudes from the internal gear **85**. As for the second planet cage **84b**, the front side portion **84f** protruding from the internal gear **85** is rotatably supported by the main body part **10** via a bearing **86**.

In addition, as for the second planet cage **84b**, a rear side portion **84r** that is the other side along the axis direction is located inside the internal gear **85**, and the rear side portion **84r** is supported to the internal gear **85** by a support member **87**. Since the internal gear **85** is fixed to the main body part **10**, the rear side portion **84r** of the second planet cage **84b** is supported by the main body part **10** via the support member **87** configuring a sliding bearing and the internal gear **85**. Note that, the support member **87** may be configured by a bearing.

Further, the decelerator **81** includes a gear holder **88** between the first planet cage **84a** and the second planetary gear **83b**. The gear holder **88** is configured by a disk-shaped member having a hole perforated at a center into which the second sun gear **82b** is inserted, and is inserted between the first planet cage **84a** and the second planetary gear **83b** outside the second sun gear **82b**, thereby securing a gap between the first planet cage **84a** and the second planetary gear **83b**.

Thereby, the second planet cage **84b** is supported at the front side portion **84f** and the rear side portion **84r** along the axis direction by the main body part **10**. Therefore, with a simple configuration, the second planet cage **84b** is suppressed from being inclined with respect to the axis direction, and changes in meshes between the sun gear and the planetary gear and between the planetary gear and the internal gear, and interferences between gears aligned in parallel in the axis direction, between a gear and a planet cage, and the like are suppressed.

Note that, like the decelerator **81** of a modified embodiment shown in FIGS. **5C** and **5D**, the gear holder **88a** may be provided integrally with the first planet cage **84a**. The gear holder **88a** is configured such that a disk-shaped member having a hole perforated at a center into which the second sun gear **82b** is inserted is provided integrally with the first planet cage **84a** outside the second sun gear **82b**. Thereby, the gear holder **88a** is inserted between the first planet cage **84a** and the second planetary gear **83b** outside the second sun gear **82b**, thereby securing a gap between the first planet cage **84a** and the second planetary gear **83b**.

Example of Embodiment of Curl Forming Unit

FIGS. **6A** to **6D** are plan views showing an example of the curl forming unit of the present embodiment. Next, an example of the curl forming unit of the present embodiment will be described with reference to each drawing.

The curl forming unit **5** includes a guide groove **52** configuring a feeding path of the wire **W** in the curl forming unit **5**, and a first guide member **53a** and a second guide

member **53b**, which are configured to curl the wire **W** in cooperation with the guide groove **52**.

The first guide member **53a** is provided on an introduction part side of the curl guide **50** for the wire **W** that is fed in the forward direction by the wire feeding unit **3**, and is arranged on a radially inner side of the loop **Ru** formed by the wire **W** with respect to the feeding path of the wire **W** by the guide groove **52**. The first guide member **53a** is configured to regulate the feeding path of the wire **W** so that the wire **W** fed along the guide groove **52** does not enter the radially inner side of the loop **Ru** formed by the wire **W**.

The second guide member **53b** is provided on a discharge part side of the curl guide **50** for the wire **W** that is fed in the forward direction by the wire feeding unit **3**, and is arranged on a radially outer side of the loop **Ru** formed by the wire **W** with respect to the feeding path of the wire **W** by the guide groove **52**.

The curl forming unit **5** includes a retraction mechanism **54** configured to retract the first guide member **53a** from the feeding path of the wire **W**. The retraction mechanism **54** is attached to a frame **55** for fixing the curl guide **50** to the main body part **10** so as to be rotatable about a shaft **54a** as a fulcrum, and is configured to be displaced in directions in which the first guide member **53a** protrudes and retracts with respect to the feeding path of the wire **W**.

The retraction mechanism **54** is urged by an urging member **56** such as a spring, in the direction in which the first guide member **53a** protrudes to the feeding path of the wire **W**.

In addition, the retraction mechanism **54** includes an induction part **57** configured to displace the retraction mechanism **54** in the direction in which the first guide member **53a** retracts with respect to the feeding path of the wire **W**. The induction part **57** is configured by an inclined surface configured, in an operation of winding the wire **W** on the reinforcing bars **S**, to be pushed by the wire **W**, thereby generating a force for displacing the retraction mechanism **54** in the direction in which the first guide member **53a** retracts with respect to the feeding path of the wire **W**.

In addition, the retraction mechanism **54** includes a wire guide part **58** configuring a part of the guide groove **52**. When the retraction mechanism **54** is moved in the direction in which the first guide member **53a** protrudes with respect to the feeding path of the wire **W**, the wire guide part **58** protrudes to the feeding path of the wire **W**, and configures a part of the guide groove **52**. In addition, when the retraction mechanism **54** is moved in the direction in which the first guide member **53a** retracts with respect to the feeding path of the wire **W**, the wire guide part **58** protrudes to the feeding path of the wire **W**, and closes a path along which the wire **W** is exposed to an outside of the guide groove **52**.

The curl forming unit **5** includes a feeding regulation part **59** against which a tip end of the wire **W** is butted, on the feeding path of the wire **W** that is curled by the curl guide **50** and guided to the binding unit **7** by the induction guide **51**.

The retraction mechanism **54** includes an opening/closing regulation portion **54b** configured to engage with the moving member **75** configured to move in conjunction with the sleeve **71** and to be in contact with an opening/closing regulation member **55a** configured to operate in conjunction with the moving member **75**. The opening/closing regulation portion **54b** comes in contact with the opening/closing regulation member **55a** in a state in which the retraction mechanism **54** has moved in the direction in which the first guide member **53a** protrudes to the feeding path of the wire

21

W, so that the rotation of the retraction mechanism **54** about the shaft **54a** as a fulcrum is regulated.

In addition, when the opening/closing regulation member **55a** is moved in conjunction with the operation of the binding unit **7** for locking the wire **W** with the locking member **70**, and an opening portion **55b** of the opening/closing regulation member **55a** is moved to a position where it faces the opening/closing regulation portion **54b** of the retraction mechanism **54**, the opening/closing regulation portion **54b** enters the opening portion **55b**, so that the regulation of rotation of the retraction mechanism **54** about the shaft **54a** as a fulcrum is released. Thereby, the retraction mechanism **54** can be moved by the rotating operation about the shaft **54a** as a fulcrum, in the direction in which the first guide member **53a** retracts with respect to the feeding path of the wire **W**.

<Example of Operation of Reinforcing Bar Binding Machine of Present Embodiment>

Subsequently, an operation of binding the reinforcing bars **S** with the wire **W** by the reinforcing bar binding machine **1A** of the present embodiment will be described with reference to each drawing.

The reinforcing bar binding machine **1A** is in a standby state where the wire **W** is sandwiched between the pair of feeding gears **30** and the tip end of the wire **W** is located between a sandwiched position by the feeding gears **30** and the fixed blade part **60** of the cutting unit **6**. Also, when the reinforcing bar binding machine **1A** is in the standby state, the sleeve **71** and the first side hook **70R**, the second side hook **70L** and the center hook **70C** attached to the sleeve **71** are moved in the rear direction denoted with the arrow **A2**, and as shown in FIG. **3A**, the first side hook **70R** is opened with respect to the center hook **70C**, and the second side hook **70L** is opened with respect to the center hook **70C**.

When the reinforcing bars **S** are inserted between the curl guide **50** and the induction guide **51** of the curl forming unit **5** and a trigger **12** is operated, the feeding motor (not shown) is driven in the forward rotation direction, so that the wire **W** is fed in the forward direction denoted with the arrow **F** by the wire feeding unit **3A**.

In a configuration where a plurality of, for example, two wires **W** are fed, the two wires **W** are fed aligned in parallel along an axis direction of the loop **Ru**, which is formed by the wires **W**, by the wire guide **4**.

The wire **W** fed in the forward direction passes between the center hook **70C** and the first side hook **70R**, and is then fed to the curl guide **50** of the curl forming unit **5**. The wire **W** passes through the curl guide **50** and is thus curled to be wound around the reinforcing bars **S**.

The wire **W** curled by the curl guide **50** is guided to the induction guide **51** and is further fed in the forward direction by the wire feeding unit **3A**, so that the wire is guided between the center hook **70C** and the second side hook **70L** by the induction guide **51**. Then, the wire **W** is fed until the tip end is butted against the feeding regulation part **59**. When the wire **W** is fed to a position at which the tip end is butted against the feeding regulation part **59**, the drive of the feeding motor (not shown) is stopped.

After stopping the feeding of the wire **W** in the forward direction, the motor **80** is driven in the forward rotation direction. In the first operation area where the wire **W** is locked by the locking member **70**, the rotation regulation blade **74a** is locked, so that the rotation of the sleeve **71** in conjunction with the rotation of the rotary shaft **72** is regulated. Thereby, the rotation of the motor **80** is converted into linear movement, so that the sleeve **71** is moved in the forward direction denoted with the arrow **A1**.

22

When the sleeve **71** is moved in the forward direction denoted with the arrow **A1**, the first side hook **70R** and the second side hook **70L** of the locking member **70** are moved toward the center hook **70C** by the rotating operations about the shaft **71b** as a fulcrum, due to the locus of the opening/closing pin **71a** and the shape of the opening/closing guide holes **73R** and **73L**.

That is, when the sleeve **71** is moved in the forward direction denoted with the arrow **A1**, the inner wall surface of the first side hook **70R** with respect to the direction in which the first side hook **70R** is closed is pushed by the opening/closing pin **71a**, in the opening/closing portion **73a** formed in the opening/closing guide hole **73R**. Thereby, the first side hook **70R** is rotated about the shaft **71b** as a fulcrum and is moved toward the center hook **70C**.

In addition, when the sleeve **71** is moved in the forward direction denoted with the arrow **A1**, the inner wall surface of the second side hook **70L** with respect to the direction in which the second side hook **70L** is closed is pushed by the opening/closing pin **71a**, in the opening/closing portion **73a** formed in the opening/closing guide hole **73L**. Thereby, the second side hook **70L** is rotated about the shaft **71b** as a fulcrum and is moved toward the center hook **70C**.

Thereby, the first side hook **70R** and the second side hook **70L** are closed with respect to the center hook **70C**.

When the first side hook **70R** is closed with respect to the center hook **70C**, the wire **W** sandwiched between the first side hook **70R** and the center hook **70C** is locked in such a manner that the wire can move between the first side hook **70R** and the center hook **70C**.

On the other hand, when the second side hook **70L** is closed with respect to the center hook **70C**, the wire **W** sandwiched between the second side hook **70L** and the center hook **70C** is locked in such a manner that the wire cannot come off between the second side hook **70L** and the center hook **70C**, within the range in which the opening/closing pin **71a** is located at the locking portion **73b** of the opening/closing guide hole **73L**, as shown in FIG. **3B**.

After advancing the sleeve **71** to a position, at which the opening/closing pin **71a** is located at the locking portion **73b** of the opening/closing guide hole **63L** and the wire **W** is locked, by the closing operation of the first side hook **70R** and the second side hook **70L**, the rotation of the motor **80** is temporarily stopped and the feeding motor (not shown) is driven in the reverse rotation direction.

Thereby, the pair of feeding gears **30** is reversely rotated and the wire **W** sandwiched between the pair of feeding gears **30** is fed in the reverse direction denoted with the arrow **R**. Since the tip end side of the wire **W** is locked in such a manner that the wire does not come off between the second side hook **70L** and the center hook **70C**, the wire **W** is wound on the reinforcing bars **S** by the operation of feeding the wire **W** in the reverse direction.

In addition, in the operation of winding the wire **W** on the reinforcing bars **S**, the induction part **57** of the retraction mechanism **54** is pushed by the wire **W**, so that the first guide member **53a** retracts with respect to the feeding path of the wire **W**.

After the wire **W** is wound on the reinforcing bars **S** and the drive of the feeding motor (not shown) in the reverse rotation direction is stopped, the motor **80** is driven in the forward rotation direction, so that the sleeve **71** is further moved in the forward direction denoted with the arrow **A1**.

FIGS. **7A** to **7G** are operation explanatory diagrams showing an example of the operations of the binding unit, the transmission unit and the cutting unit according to the present embodiment. As shown in FIG. **7A**, when the sleeve

23

71 is moved in the forward direction denoted with the arrow A1, the moving member 75 is moved in the forward direction denoted with the arrow A1 in conjunction with the sleeve 71.

As shown in FIG. 7B, the engaging portion 75a is engaged with the engaged portion 93 of the cam 90 by the operation of the moving member 75 moving in the forward direction denoted with the arrow A1. A region from when the sleeve 71 is moved in the forward direction denoted with the arrow A1 until the engaging portion 75a of the moving member 75 is engaged with the engaged portion 93 of the cam 90 is referred to as an idle running region.

When the moving member 75 is further moved in the forward direction denoted with the arrow A1, the engaged portion 93 is pushed forward, so that the cam 90 is rotated in the direction of the arrow C1 about the shaft 90a as a fulcrum. When the cam 90 is rotated in the direction of the arrow C1, a portion of the cam groove 92 intersecting the guide portion 10b changes, and the length from the shaft 90a of the cam 90 to the intersection of the cam groove 92 and the guide portion 10b changes in an increasing direction.

As for the link 91, the shaft portion 91a is inserted into the cam groove 92 and the guide portion 10b at the intersection of the cam groove 92 and the guide portion 10b, and the rotating operation of the cam 90 about the shaft 90a as a fulcrum moves the shaft portion 91a along the cam groove 92 and the guide portion 10b.

Thereby, when the cam 90 is rotated in the direction of the arrow C1 and the length from the shaft 90a of the cam 90 to the intersection of the cam groove 92 and the guide portion 10b changes in an increasing direction, the shaft portion 91a of the link 91 is moved along the cam groove 92 and the guide portion 10b, so that the shaft portion 91a is moved in the direction away from the shaft 90a of the cam 90.

As for the transmission unit 9, when the shaft portion 91a of the link 91 is moved in the direction away from the shaft 90a of the cam 90, the rotating operation of the cam 90 is converted into movement along the extension direction of the link 91.

Thereby, the rotating operation of the cam 90 is transmitted to the movable blade part 61 via the link 91, so that the movable blade part 61 is rotated in the direction of the arrow D1.

When the movable blade part 61 is rotated in the direction of the arrow D1, one wire W of the two wires W aligned in parallel is pressed against the end edge portion of the first butting portion 60b of the fixed blade part 60 by the operation of the movable blade part 61, and the other wire W enters the second butting portion 60c of the fixed blade part 60, so that the cutting of the one wire W is started prior to the other wire W.

A region from when the cam 90 is rotated in the direction of the arrow C1 about the shaft 90a as a fulcrum, so that the movable blade 61 is rotated in the direction of the arrow D1 until the cutting of the first wire W by the movable blade part 61 is started, as shown in FIG. 7C, is referred to as an idling region. The idle running region and the idling region are regions in which a load that is applied to the movable blade part 61 is low.

In the idling region, the first range 92a of the cam groove 92 intersects the guide portion 10b. While the first range 92a of the cam groove 92 intersects the guide portion 10b, the length from the shaft 90a to the intersection of the cam groove 92 and the guide portion 10b is shorter and the amount of change in length between the shaft 90a and the cam groove 92 becomes larger, as compared with the case where the second range 92b intersects the guide portion 10b.

24

Thereby, the amount of rotation of the movable blade part 61 becomes relatively large with respect to the amount of movement of the sleeve 71 that rotates the cam 90. On the other hand, in the idling region, since the cutting of the wire W has not been started, there is no wire cutting load that is applied to the movable blade part 61, so that an increase in load that is applied to the cam 90 connected to the movable blade part 61 via the link 91 is suppressed.

Since the cam 90 is connected to the sleeve 71 via the moving member 75, the increase in load that is applied to the cam 90 is suppressed, so that an increase in load that is applied to the rotary shaft 72 that moves the sleeve 71 and to the motor 80 connected to the rotary shaft 72 via the decelerator 81 is suppressed.

Therefore, in the region in which the load is low until the cutting of the first wire W is started, a time consumed to rotate the movable blade part 61 to a position where the cutting of the wire W is started can be shortened by relatively increasing the amount of rotation of the movable blade part 61.

When the moving member 75 is moved in the forward direction denoted with the arrow A1 to the position where the movable blade part 61 starts cutting of the first wire W, the cam 90 rotates about the shaft 90a as a fulcrum, as shown in FIG. 7D, so that the second range 92b of the cam groove 92 intersects the guide portion 10b.

While the second range 92b of the cam groove 92 intersects the guide portion 10b, the length from the shaft 90a of the cam 90 to the intersection of the cam groove 92 and the guide portion 10b changes in an increasing direction, and the shaft portion 91a of the link 91 is moved along the cam groove 92 and the guide portion 10b, so that the shaft portion 91a is moved in the direction away from the shaft 90a of the cam 90.

Thereby, the moving member 75 is further moved in the forward direction denoted with the arrow A1 to rotate the cam 90 in the direction of the arrow C1, and the rotating operation of the cam 90 is transmitted to the movable blade part 61 via the link 91, so that the movable blade part 61 is further rotated in the direction of the arrow D1 to start cutting of the first wire W.

After the movable blade part 61 is rotated in the direction of the arrow D1 to start cutting of the first wire W, which is one wire, when the first wire W is cut to a predetermined position, the second wire W, which is the other wire, is pressed against the end edge portion of the second butting portion 60c of the fixed blade part 60 by the operation of the movable blade part 61.

Thereby, cutting of the second wire W is started. In the present example, after starting the cutting of the first wire W, when the first wire W is cut in half or more in the radial direction, the cutting of the second wire W is started.

As described above, while the cutting of the first wire W is started and the second range 92b of the cam groove 92 intersects the guide portion 10b, the length from the shaft 90a to the intersection of the cam groove 92 and the guide portion 10b is longer and the amount of change in length between the shaft 90a and the cam groove 92 becomes smaller, as compared with the case where the first range 92a intersects the guide portion 10b.

Thereby, the amount of rotation of the movable blade part 61 becomes relatively small with respect to the amount of movement of the sleeve 71. On the other hand, the force that can be generated by the movable blade part 61 by operating the movable blade part 61 with the cam 90 via the link 91 increases.

25

When the cutting of the first wire W is started, the load that is applied to the movable blade part 61 increases. On the other hand, the force that can be generated by the movable blade part 61 increases, so that the load that is applied to the movable blade part 61 is canceled and the increase in load that is applied to the cam 90 connected to the movable blade part 61 via the link 91 is suppressed.

The increase in load that is applied to the cam 90 is suppressed, so that an increase in load that is applied to the rotary shaft 72 that moves the sleeve 71 and to the motor 80 connected to the rotary shaft 72 via the decelerator 81 is suppressed.

When the movable blade part 61 is rotated in the direction of the arrow D1 and the moving member 75 is moved in the forward direction denoted with the arrow A1 from the position where the cutting of the first wire W is started to the position where the cutting of the second wire W is started, the cam 90 is rotated about the shaft 90a as a fulcrum, as shown in FIG. 7E, so that the second range 92b of the cam groove 92 intersects the guide portion 10b.

When the movable blade part 61 is further rotated in the direction of the arrow D1, the cutting of the one wire W for which cutting has been started first is completed. When the movable blade part 61 is further rotated in the direction of the arrow D1, the cutting of the other wire W for which cutting has been started later is completed.

When the movable blade part 61 is rotated in the direction of the arrow D1 and the moving member 75 is moved in the forward direction denoted with the arrow A1 from a position where the cutting of the second wire W is started to a position where the cutting of the second wire W ends, as described above, the cam 90 is rotated about the shaft 90a as a fulcrum, as shown in FIG. 7F, so that the second range 92b of the cam groove 92 intersects the guide portion 10b.

When the cutting of the second wire W is started, the load that is applied to the movable blade part 61 further increases. On the other hand, the force that can be generated by the movable blade part 61 increases, so that the load that is applied to the movable blade part 61 is canceled and the increase in load that is applied to the cam 90 connected to the movable blade part 61 via the link 91 is suppressed.

The increase in load that is applied to the cam 90 is suppressed, so that an increase in load that is applied to the rotary shaft 72 that moves the sleeve 71 and to the motor 80 connected to the rotary shaft 72 via the decelerator 81 is suppressed.

Therefore, in a region in which the load is high from when the cutting of the first wire W is started until the cutting of the second wire W ends, the increase in load that is applied to the motor 80 can be suppressed by increasing the force that can be generated by the movable blade part 61. In addition, in the region in which the load is high, the amount of rotation of the movable blade part 61 becomes relatively small, but in the region in which the load is low, the time consumed until the cutting of the wire W ends can be suppressed from lengthening by relatively increasing the amount of rotation of the movable blade part 61.

When the moving member 75 is moved in the forward direction denoted with the arrow A1 to the position where the movable blade part 61 ends the cutting of the second wire W, the cam 90 is rotated about the shaft 90a as a fulcrum, as shown in FIG. 7G, so that the third range 92c of the cam groove 92 intersects the guide portion 10b.

While the third range 92c of the cam groove 92 intersects the guide portion 10b, the length from the shaft 90a to the intersection of the cam groove 92 and the guide portion 10b is substantially equivalent and the amount of change in

26

length between the shaft 90a and the cam groove 92 is further smaller and becomes substantially constant, as compared with the case where the second range 92b intersects the guide portion 10b.

Thereby, the relative amount of rotation of the movable blade part 61 becomes smaller with respect to the amount of movement of the sleeve 71. When the cutting of the wire W ends, it is not necessary to rotate the movable blade part 61. On the other hand, after the cutting of the wire W, in order to bend the wire W, the sleeve 71 needs to be moved in the forward direction denoted with the arrow A1.

Therefore, while the third range 92c of the cam groove 92 intersects the guide portion 10b, the amount of rotation of the movable blade part 61 is reduced with respect to the amount of movement of the sleeve 71, and the increase in load due to the rotation of the movable blade part 61 after the cutting of the wire W is suppressed, so that the increase in load that is applied to the cam 90 connected to the movable blade part 61 via the link 91 is suppressed.

Therefore, in the region from when the cutting of the second wire W ends until the movement of the sleeve 71 is stopped, the increase in load that is applied to the cam 90 due to the rotation of the movable blade part 61 is suppressed, so that the increase in load that is applied to the rotary shaft 72 that moves the sleeve 71 and to the motor 80 connected to the rotary shaft 72 via the decelerator 81 can be suppressed.

Note that, the amount of movement of the sleeve 71 per rotation of the rotary shaft 72 is defined by a lead angle of the feeding screw 72a. Therefore, the lead angle of the feeding screw 72a is increased with respect to the reinforcing bar binding machine of the related art. On the other hand, in the region in which the load that is applied to the movable blade part 61 is high, the amount of rotation of the movable blade part 61 becomes relatively small, but the force that can be generated by the movable blade part 61 is increased, and in the region in which the load that is applied to the movable blade part 61 is low, the amount of rotation of the movable blade part 61 is relatively increased. Thereby, the time consumed until the cutting of the wire W ends can be suppressed from lengthening, and a time required for the whole binding operation can be shortened, as compared with the related art.

Further, in the operation of cutting the wire W whose cross-sectional shape is circular, the load becomes highest immediately before the wire that the blade part has reached a position of a diameter is cut. Therefore, in the configuration where the two wires W aligned in parallel are cut, a phase difference is provided for timings at which the cuttings of the wires W are started. First, after starting the cutting of the first wire W, when the wire W is cut to a position of a half or more in the radial direction, the cutting of the second wire W is started.

As compared with a case where two wires W aligned in parallel are cut at the same time, cutting one wire W reduces the load. Thereby, the load is reduced by starting the cutting of one wire W in advance. In addition, after the first wire W is cut to the position of a half or more in the radial direction and therefore the position where the load is the highest is passed, the cutting of the second wire W is started. Thereby, even when the two wires W are cut, the load is reduced. Further, the cutting of the second wire W is started before the cutting of the first wire W is completed. Thereby, an increase in time required for the cutting is suppressed.

Further, when the sleeve 71 is moved in the forward direction denoted with the arrow A1 by the operation of cutting the wire W wound on the reinforcing bars S, and as shown in FIG. 3C, the opening/closing pin 71a is moved to

27

the range in which it is located at the unlocking portion 73c of the opening/closing guide hole 73L, the second side hook 70L becomes movable in the direction away from the center hook 70C by a predetermined amount.

As described above, in the operation of feeding the wire W in the reverse direction and winding the wire on the reinforcing bars S, the tip end side of the wire W needs to be locked in such a manner that the wire does not come off between the second side hook 70L and the center hook 70C. On the other hand, a reactive force of the force for pressing the wire W against the center hook 70C with the second side hook 70L is applied to the sleeve 71, and this reactive force becomes the load that is applied to the rotary shaft 72 that moves and rotates the sleeve 71 and to the motor 80 connected to the rotary shaft 72 via the decelerator 81.

Therefore, the second side hook 70L is provided with the locking portion 73b and the unlocking portion 73c in the opening/closing guide hole 73L, and in the operation of winding the wire W on the reinforcing bars S, the sleeve 71 is moved to the position where the opening/closing pin 71a faces the locking portion 73b of the opening/closing guide hole 73L, and after the wire W is wound on the reinforcing bars S, the sleeve 71 is moved to the position where the opening/closing pin 71a faces the unlocking portion 73c of the opening/closing guide hole 73L.

Thereby, in the operation of winding the wire W on the reinforcing bars S, the tip end side of the wire W can be locked in such a manner that the wire does not come off between the second side hook 70L and center hook 70C. In addition, after winding the wire W on the reinforcing bars S, the second side hook 70L becomes movable in the direction away from the center hook 70C by a predetermined amount, the reactive force of the force of pressing the wire W against the center hook 70C with the second side hook 70L is reduced, and the load that is applied to the motor 80 is reduced.

By driving the motor 80 in the forward rotation direction, the sleeve 71 is moved in the forward direction denoted with the arrow A1, so that the bent portions 71c1 and 71c2 are moved toward the reinforcing bars S almost simultaneously with the cutting of the wire W as described above. Thereby, the tip end side of the wire W locked by the center hook 70C and the second side hook 70L is pressed toward the reinforcing bars S and bent toward the reinforcing bars S at the locking position as a fulcrum by the bending portion 71c1. The sleeve 71 is further moved in the forward direction, so that the wire W locked between the second side hook 70L and the center hook 70C is maintained sandwiched by the bending portion 71c1.

In addition, the terminal end side of the wire W locked by the center hook 70C and the first side hook 70R and cut by the cutting unit 6 is pressed toward the reinforcing bars S and bent toward the reinforcing bars S at the locking position as a fulcrum by the bending portion 71c2. The sleeve 71 is further moved in the forward direction, so that the wire W locked between the first side hook 70R and the center hook 70C is maintained sandwiched by the bending portion 71c2.

After bending the tip end side of the wire W and the terminal end side after the cutting toward the reinforcing bars S, the motor 80 is further driven in the forward rotation direction, so that the sleeve 71 is further moved in the forward direction. When the sleeve 71 is moved to a predetermined position and therefore reaches the operation region in which the wire W locked by the locking member 70 is twisted, the locking of the rotation regulation blade 74a is released.

28

Thereby, the motor 80 is further driven in the forward rotation direction, so that the sleeve 71 is rotated in conjunction with the rotary shaft 72 and the wire W locked by the locking member 70 is twisted.

In the second operation region in which the sleeve 71 is rotated to twist the wire W, the binding unit 7 twists the wire W locked by the locking member 70, so that a force of pulling the sleeve 71 forward along the axis direction of the rotary shaft 72 is applied. On the other hand, when a force to move the sleeve 71 forward along the axis direction is applied, the rotary shaft 72 moves forward while receiving a force pushed backward by the spring 72c, and twists the wire W while moving forward.

Therefore, the wire W is twisted while the locking member 70, the sleeve 71, and the rotary shaft 72 are moved forward with receiving the force pushed backward by the spring 72c, and therefore, a gap between the twisted portion of the wire W and the reinforcing bar S becomes small and the wire is brought into close contact with the reinforcing bar S along the reinforcing bar S. Thereby, the slack before twisting the wire W can be removed, and the reinforcing bars S can be bound in a state where the wire W is in close contact with the reinforcing bars S.

When it is detected that the load that is applied to the motor 80 is maximized as the wire W is twisted, the forward rotation of the motor 80 is stopped. Next, when the motor 80 is driven in the reverse rotation direction, the rotary shaft 72 is reversely rotated and the sleeve 71 is reversely rotated in conjunction with the reverse rotation of the rotary shaft 72, the rotation regulation blade 74a is locked, so that the rotation of the sleeve 71 in conjunction with the rotation of the rotary shaft 72 is regulated. Thereby, the sleeve 71 is moved in the direction of the arrow A2, which is a backward direction.

When the sleeve 71 is moved in the backward direction, the bending portions 71c1 and 71c2 are away from the wire W, and the holding of the wire W by the bending portions 71c1 and 71c2 is released. In addition, when the sleeve 71 is moved in the backward direction, the opening/closing pin 71a passes through the opening/closing guide holes 73R and 73L. Thereby, the first side hook 70R is moved away from the center hook 70C by the rotating operation about the shaft 71b as a fulcrum. In addition, the second side hook 70L is moved away from the center hook 70C by the rotating operation about the shaft 71b as a fulcrum. Thereby, the wire W comes off from the locking member 70.

Note that, as in the opening/closing guide hole 73L of the modified embodiments shown in FIGS. 3D to 3F, in the configuration where the opening/closing guide hole 73L is provided with the second locking portion 73d, when the sleeve 71 is further moved in the forward direction to a position where the operation of twisting the wire W becomes possible, the opening/closing pin 71a is located at the second locking portion 73d of the opening/closing guide hole 73L. Thereby, even when the force by which the wire W is twisted is applied to the wire W, the wire W is suppressed from coming off between the second side hook 70L and the center hook 70C.

Modified Embodiment of Implementation of Transmission Unit

FIGS. 8A to 8C are side views showing a modified embodiment of the transmission unit of the present embodiment, and FIGS. 9A to 9C are side cross-sectional views showing the modified embodiment of the transmission unit of the present embodiment. Next, a transmission unit 9B of the modified embodiment of the present embodiment will be described with reference to each drawing.

The transmission unit 9B includes a cutter lever 95 configured to rotate by an operation of the binding unit 7, and a link 91 configured to connect the cutter lever 95 and the movable blade part 61. The transmission unit 9B is configured to transmit an operation of the binding unit 7 to the cutter lever 95 and the movable blade part 61 of the cutting unit 6 via the link 91.

The transmission unit 9B is supported so that the cutter lever 95 can rotate about the shaft 90b as a fulcrum. The shaft 90b is attached to the frame 10a attached to the inside of the main body part 10.

The cutter lever 95 is an example of the displacement member, and includes a first cutter lever 95a and a second cutter lever 95b connected to the sleeve 71 via the moving member 75. The cutter lever 95 is configured such that the first cutter lever 95a is engaged with the first engaging portion 75b provided to the moving member 75 and the second cutter lever 95b is engaged with the second engaging portion 75c provided to the moving member 75.

The cutter lever 95 is configured such that a length from an action point, which is the second connection portion connected to the sleeve 71, to be pushed by the moving member 75 configured to move in conjunction with the sleeve 71 to the shaft 90b is different in the first cutter lever 95a and the second cutter lever 95b. The length from the shaft 90b to the action point to be pushed by the moving member 75 is configured to be longer in the second cutter lever 95b than in the first cutter lever 95a.

That is, the length from the second engaging portion 75c, which is the action point to be pushed by the moving member 75 in the second cutter lever 95b, to the shaft 90b is configured to be greater than the length from the first engaging portion 75b, which is the action point to be pushed by the moving member 75 in the first cutter lever 95a, to the shaft 90b.

When the moving member 75 is moved in the forward direction in conjunction with the sleeve 71 moving in the forward direction denoted with the arrow A1, first, the first engaging portion 75b is engaged with the first cutter lever 95a. When the sleeve 71 is further moved in the forward direction denoted with the arrow A1, the second engaging portion 75c is engaged with the second cutter lever 95b. Further, the engagement between the first cutter lever 95a and the first engaging portion 75b is released.

As for the link 91, an end portion in the forward direction denoted with the arrow A1 is connected to the movable blade part 61, and an end portion in the backward direction denoted with the arrow A2 is connected to the cutter lever 95.

Next, operations of the transmission unit 9B are described. When the sleeve 71 is moved in the forward direction denoted with the arrow A1, the moving member 75 is moved in the forward direction denoted with the arrow A1 in conjunction with the sleeve 71. As shown in FIG. 9B, the first engaging portion 75b is engaged with the first cutter lever 95a by the moving operation of the moving member 75 in the forward direction denoted with the arrow A1.

When the moving member 75 is further moved in the forward direction denoted with the arrow A1, the cutter lever 95 is rotated in the direction of the arrow C1 about the shaft 90b as a fulcrum with a ratio corresponding to the length from the shaft 90b to the action point pushed by the first engaging portion 75b of the moving member 75 in the first cutter lever 95a with respect to the amount of movement of the sleeve 71.

When the cutter lever 95 is rotated in the direction of the arrow C1, the rotating operation of the cutter lever 95 is

transmitted to the movable blade part 61 via the link 91, so that the movable blade part 61 is rotated in the direction of the arrow D1. Therefore, the movable blade part 61 is rotated in the direction of the arrow D1 by the moving operation of the sleeve 71 in the forward direction, so that cutting of the wire W is started.

When the sleeve 71 is further moved in the forward direction denoted with the arrow A1, the second engaging portion 75c of the moving member 75 is engaged with the second cutter lever 95b, as shown in FIG. 8C. Thereby, the cutter lever 95 is rotated in the direction of the arrow C1 about the shaft 90b as a fulcrum with a ratio corresponding to the length from the shaft 90b to action point pushed by the second engaging portion 75c of the moving member 75 in the second cutter lever 95b with respect to the amount of movement of the sleeve 71. Further, the engagement between the first cutter lever 95a and the first engaging portion 75b is released.

The duration for which the first cutter lever 95a and the first engaging portion 75b are engaged is a duration from when the movable blade part 61 starts rotation in the cutting unit 6 until the cutting of the first wire W is started. In addition, the duration for which the second cutter lever 95b and the second engaging portion 75c are engaged is a duration from when the movable blade part 61 is further rotated in the cutting unit 6 and the cutting of the first wire W is started until the cutting of the second wire W ends.

The cutter lever 95 is configured such that the length from the shaft 90b to the action point pushed by the moving member 75 is longer in the second cutter lever 95b than in the first cutter lever 95a. Thereby, while the first cutter lever 95a and the first engaging portion 75b are engaged, the amount of rotation of the movable blade part 61 becomes relatively large with respect to the amount of movement of the sleeve 71 that rotates the cutter lever 95.

On the other hand, since the cutting of the wire W is not started while the first cutter lever 95a and the first engaging portion 75b are engaged, the increase in load that is applied to the movable blade part 61 is suppressed, and the increase in load that is applied to the cutter lever 95 connected to the movable blade part 61 via the link 91 is suppressed.

Since the cutter lever 95 is connected to the sleeve 71 via the moving member 75, the increase in load that is applied to the cutter lever 95 is suppressed, so that the increase in load that is applied to the rotary shaft 72 that moves the sleeve 71 and to the motor 80 connected to the rotary shaft 72 via the decelerator 81 is suppressed.

Therefore, in the region in which the load is low until the cutting of the first wire W is started, a time consumed to rotate the movable blade part 61 to a position where the cutting of the wire W is started can be shortened by relatively increasing the amount of rotation of the movable blade part 61.

While the second cutter lever 95b and the second engaging portion 75c are engaged, the amount of rotation of the movable blade part 61 becomes relatively small with respect to the amount of movement of the sleeve 71 that rotates the cutter lever 95. On the other hand, since the length from the shaft 90b to the action point pushed by the moving member 75 is configured to be longer in the second cutter lever 95b than in the first cutter lever 95a, the force that can be generated by the movable blade part 61 from the cutter lever 95 via the link 91 increases.

When the cutting of the first wire W is started, the load that is applied to the movable blade part 61 increases. On the other hand, the force that can be generated by the movable blade part 61 increases, so that the load that is applied to the

31

movable blade part **61** is canceled and the increase in load that is applied to the cutter lever **95** connected to the movable blade part **61** via the link **91** is suppressed.

The increase in load that is applied to the cutter lever **95** is suppressed, so that the increase in load that is applied to the rotary shaft **72** that moves the sleeve **71** and to the motor **80** connected to the rotary shaft **72** via the decelerator **81** is suppressed.

Therefore, in a region in which the load is high from when the cutting of the first wire **W** is started until the cutting of the second wire **W** ends, the increase in load that is applied to the motor **80** can be suppressed by increasing the force that can be generated by the movable blade part **61**. In addition, in the region in which the load is high, the amount of rotation of the movable blade part **61** becomes relatively small, but in the region in which the load is low, the time consumed until the cutting of the wire **W** ends can be suppressed from lengthening by relatively increasing the amount of rotation of the movable blade part **61**.

Note that, in the above embodiment, the cutter lever **75** has such a configuration that whether the first engaging portion **75b** of the moving member **75** and the first cutter lever **95a** are engaged or whether the second engaging portion **75c** of the moving member **75** and the second cutter lever **95b** are engaged is switched by the rotating operation of the cutter lever **85** about the shaft **90b** as a fulcrum, and therefore, the length from the shaft **90b** to the first connection portion connected to the sleeve **71** is switched.

Thereby, the cutter lever **95** makes it possible to switch the amount of rotation (amount of movement) of the movable blade part **61** and the force that can be generated by the movable blade part **61**, within the rotating range (moving range) of the movable blade part **61**.

On the other hand, the cutter lever **95** may have such a configuration that the portion to which the link **91** is connected can be switched by the rotating operation of the cutter lever **85** about the shaft **90b** as a fulcrum, and therefore, the length from the shaft **90b** to the second connection portion connected to the link **91** can be switched.

What is claimed is:

1. A binding machine comprising:

a wire feeding unit comprising a pair of gears configured to feed a wire;

a curl forming unit comprising a curl guide configured to form a path along which the wire fed by the wire feeding unit is to be wound around an object;

a cutting unit comprising blade parts configured to cut the wire wound on the object;

a binding unit including a locking member configured to lock the wire, a sleeve configured to actuate the locking member, and a rotary shaft configured to move and rotate the sleeve, the binding unit being configured to twist the wire wound on the object and cut by the cutting unit; and

a drive unit configured to drive the binding unit, wherein the drive unit comprises a motor and a decelerator configured to perform deceleration and amplification of torque,

wherein the decelerator comprises:

a sun gear,

a planetary gear in mesh with the sun gear,

32

a planet cage configured to support the planetary gear, and

an internal gear in mesh with the planetary gear, and wherein the planet cage includes a front side portion protruding from the internal gear along an axis direction and a rear side portion located inside the internal gear along the axis direction,

wherein the front side portion and the rear side portion of the planet cage are rotatably supported, and

wherein the rear side portion is rotatably supported on the internal gear.

2. The binding machine according to claim 1, wherein the planet cage is configured such that one side along the axis direction is rotatably supported by a main body part to which the internal gear is fixed and the other side is rotatably supported by the internal gear.

3. The binding machine according to claim 1, wherein the planet cage is configured such that one side along the axis direction is supported on an inner side of the internal gear.

4. A binding machine comprising:

a wire feeding unit comprising a pair of gears configured to feed a wire;

a curl forming unit comprising a curl guide configured to form a path along which the wire fed by the wire feeding unit is to be wound around an object;

a cutting unit comprising blade parts configured to cut the wire wound on the object;

a binding unit including a locking member configured to lock the wire, a sleeve configured to actuate the locking member, and a rotary shaft configured to move and rotate the sleeve, the binding unit being configured to twist the wire wound on the object and cut by the cutting unit; and

a drive unit configured to drive the binding unit, wherein the drive unit comprises a motor and a decelerator configured to perform deceleration and amplification of torque,

wherein the decelerator comprises:

a first sun gear attached to a shaft of the motor,

a first planetary gear in mesh with the first sun gear,

a first planet cage configured to support the first planetary gear,

a second sun gear provided to the first planet cage,

a second planetary gear in mesh with the second sun gear,

an internal gear in mesh with the second planetary gear, and

a second planet cage configured to support the second planetary gear,

wherein the second planet cage includes a front side portion protruding from the internal gear along an axis direction and a rear side portion located inside the internal gear along the axis direction,

wherein the front side portion and the rear side portion of the second planet cage are rotatably supported, and wherein the rear side portion is rotatably supported on the internal gear.

5. The binding machine according to claim 4, wherein the decelerator comprises a gear holder between the first planet cage and the second planetary gear.

* * * * *